Dilution Rules and the Value of a Most-Favored-Nation Clause

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Abstract

We ask what a most-favored-nation (MFN) clause held by an accelerator is worth to the startups that grant it, as in the $375,000 uncapped MFN SAFE that Y Combinator added to its standard deal in 2022. In our model, the accelerator and a seed investor with market power finance an experiment, and competitive Series A investors finance its scaling. The value of the clause depends on the dilution rules of the seed contract. A firm without a seed round loses, because the MFN SAFE replaces Series A money that diluted the accelerator. With a seed equity round, the clause takes part of the seed investor's claims, and the firms whose need comes close to their value lose their seed round. With a capped seed SAFE, the seed investor invests more, and only the firms whose threshold is set by the SAFE's discount lose. All the losses come from the claim: the same cash without the clause harms no firm.

JEL Codes: G24, G23, D82, D86

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1 Introduction

Most accelerators offer the same standard deal to every startup in a batch. The most common deal is a small amount for a share of equity, which Techstars offered from its first program in 2007.[1] Y Combinator (YC) put on average about $17,000 in equity for 7% and $80,000 in a safe converting at the terms of the next round.[2] In 2014, it replaced this structure by a single deal, "$120k for 7% equity (regardless of the number of founders)", to avoid complexity.[3] Sam Altman, president of YC in 2014, argued: "Speaking of hating complexity, we've tried to make the new structure really simple. The convertible notes and safes we used got complicated in terms of how they got priced, and complexity often causes unintended consequences. It was hard for founders to actually predict how much total dilution they were looking at." But, in January 2022, YC added to its $125,000 for 7% a $375,000 uncapped Simple Agreement for Future Equity (SAFE) with a most-favored-nation (MFN) clause: the SAFE converts at the best terms the startup grants to intermediate investors before its first equity round. The deal is still the same for every startup in the batch, and according to Geoff Ralston, president of YC in 2022, it should "remove the immediate pressure to fundraise and accept less than favorable terms".[4]

The MFN clause answers a structural problem of sequential venture financing. The accelerator sources and trains startups and hands them to venture capitalists (VCs) (Gonzalez-Uribe and Leatherbee, 2018; Hallen et al., 2020). When it invests, neither the startup's probability of success nor its exact financing need is known, an uncertainty that comes with early-stage financing (Nanda and Rhodes-Kropf, 2017; Lerner and Nanda, 2020; Ewens et al., 2018). Information arrives in sequence: the financing need is known before the quality (Kerr et al., 2014; Ewens et al., 2018; Bergemann and Hege, 1998, 2005). A startup from an accelerator may therefore need additional funding before it can raise an equity round (Coyle and Green, 2014), and the intermediate investors who provide it have market power over the entrepreneur. Nondiscrimination clauses such as MFN clauses are meant to protect early signatories against this opportunism, but McAfee and Schwartz (1994) show that they generally fail to do so. Ralston also announced that the new deal should help founders "raise a successful seed round".[5] This paper asks what the clause is worth to the startups that grant it: whether it helps them raise a seed round and reach Series A, and which of them it harms.

We summarize our findings. First, with equity rounds only, the share the seed investor

[1] "Initially, Techstars invested between $6,000 and $18,000 in early stage companies" (Techstars, Wikipedia).

[2] TechCrunch, April 22, 2014.

[3] Sam Altman, "The New Deal," Y Combinator blog, April 22, 2014, which is also the source of the quotation below.

[4] "YC's $500,000 Standard Deal," Y Combinator blog, January 11, 2022, `https://www.ycombinator.com/blog/ycs-500-000-standard-deal`.

[5] Same source. The MFN SAFE "will take on the terms of the lowest cap safe (or other most favorable terms)" issued "between the start of the batch and the next equity round".

takes decides which projects reach Series A. The Series A round raises the threshold one for one, and the seed investor's share raises it by diluting the entrepreneur, whose effort cost must be covered (Proposition 1). The seed investor covers the firm's need exactly: the cash it would advance beyond the need is lost on the projects that do not reach Series A (Proposition 2). More accelerator cash means more firms without a seed round and fewer liquidations, and it lowers no firm's probability of reaching Series A (Propositions 4 and 5). Second, YC's reform, \$375,000 more in cash placed in the MFN SAFE, lowers the probability of reaching Series A of the firms without a seed round, whatever the seed contract. Their threshold rises by $\rho_a^0/(1-\rho_a^0)$ times the MFN SAFE: the MFN SAFE converts at the Series A price and does not dilute the accelerator's post-money SAFE, while the Series A round it replaces did. With a seed equity round, the MFN SAFE converts at the seed price and takes part of the seed investor's claims, and the firms whose need comes within $\rho_a^0/(1-\rho_a^0)$ times the MFN SAFE of the value of these claims lose their seed round (Proposition 9). Third, with a capped seed SAFE, the MFN SAFE joins the seed investor's SAFE block. Without the clause, the seed investor covers the need exactly, whether it has market power or not (Propositions 2, 11 and 19). The clause is what makes it invest beyond the need: it holds only part of the block, and when the need is small a dollar it adds raises its part by more than the dollar costs (Proposition 11). Firms whose threshold is set by the seed investor's share receive at least as much in SAFEs and do not lose, while the firms whose threshold is set by the SAFE's discount lose when they receive more in SAFEs (Proposition 13). At every amount, an equity round gives the seed investor at least as much as a capped SAFE (Proposition 15). Split into the cash it adds and the claim it gives the accelerator, the reform has all its losers in the claim: the cash alone lowers no firm's probability of reaching Series A (Proposition 16). Fourth, the seed investor's best capped SAFE has a cap and no discount (Proposition 17). A discount without a cap, which works as an uncapped convertible note, gives it less than a cap (Proposition 18), and a discount set by the market adds to the firms that lose (Section 6.3). With competitive seed investors (Appendix A), the reform still harms the firms that need no seed round and the firms that raise a seed equity round, and it no longer harms the firms that use a capped SAFE: the MFN SAFE then replaces seed money that did not dilute the accelerator either (Proposition 21).

To arrive at these conclusions, we set up a three-stage game between risk-neutral agents. We see entrepreneurship as experimentation (Kerr et al., 2014): a startup first pays to learn what its project is worth, then pays to scale it. The cost of the experiment is a liquidity shock, unknown when the accelerator invests. The accelerator offers a fixed amount for a fixed share before the shock. The seed investor, with a limited budget, covers the shock after observing it. The Series A investor finances the remaining cost of scaling once the project's quality is public, in competition with other Series A investors, while the seed investor has market power. The entrepreneur bears a private effort cost, so the investors' shares must leave the entrepreneur enough of the outcome. Each equity round dilutes every earlier shareholder, and the accelerator's post-money SAFE is diluted by equity rounds but not by the other SAFEs. We solve the game by backward induction, first with equity rounds only (Section 3),

then with the MFN SAFE and a seed equity round (Section 4), and then with the MFN SAFE and a capped seed SAFE (Section 5). Section 6 compares three designs of the capped SAFE: a cap alone, a discount alone, and a cap with a discount set by the market. Appendix A lets seed investors compete.

Our contribution is twofold. First, we propose a tractable model of sequential venture financing in which the intermediate investor's contract, an equity round or a SAFE, decides which projects reach Series A. Second, we evaluate the MFN SAFE that accelerators adopted after YC. The clause makes seed investors invest more, but it lowers the probability of reaching Series A of the firms without a seed round, of the firms whose threshold is set by the SAFE's discount, and, with a seed equity round, of the firms that lose their seed round. Its aggregate effect, like its effect on the probability of raising a seed round, depends on the distribution of liquidity shocks (experimentation costs).

Our paper relates to four literatures.

The first studies entrepreneurship as experimentation. Kerr et al. (2014) argue that the probability of success of a new venture is unknown until someone invests in it, and that the cost of experimenting shapes who finances innovation. Ewens et al. (2018) show that when cloud computing lowered this cost, VCs started to give little money and little governance to many more startups, and to abandon more of them. Nanda and Rhodes-Kropf (2017) show that investors can protect a startup against a lack of future funding by giving it more money upfront, at the cost of the option to stop. Investors also stage their financing to learn before investing more. Gompers (1995) shows that VCs stage more often when agency costs are higher. Neher (1999) explains staging by the entrepreneur's ability to renegotiate once the money is invested, early rounds building the collateral that supports later ones. Bergemann and Hege (1998, 2005) show that funding stops too early when the learning process is subject to moral hazard. We add to this literature by modeling who finances the experiment and on which terms. The share the seed investor takes decides which experiments are followed by scaling, and the seed investor advances more than the cost of the experiment only when it shares its claims with the accelerator's MFN SAFE.

The second studies the securities used in venture capital. Cornelli and Yosha (2003) show that convertible debt deters the entrepreneur from manipulating interim results to avoid liquidation, because a better interim result also makes the VC convert. Schmidt (2003) shows that convertible securities give both the entrepreneur and the VC incentives to invest in the project. Repullo and Suarez (2004) derive the optimal securities when the project is financed in two stages under double-sided moral hazard. In Casamatta (2003), the VC both finances and advises the firm, and receives common stock when it invests little and convertible securities when it invests a lot. Hellmann (1998) explains why entrepreneurs give VCs the right to replace them. Admati and Pfleiderer (1994) show that a VC who receives a fixed fraction of the payoff and finances the same fraction of every later round has no incentive to misprice later rounds. Kaplan and Strömberg (2003, 2004) document that VC contracts allocate

cash-flow, board, voting and liquidation rights separately, and that riskier deals give the VC more of these rights. Coyle and Green (2014) document that after 2005, convertible notes, until then used for bridge loans, became a seed financing instrument with a conversion price cap, followed by the SAFE. We add to this literature by comparing three seed contracts used in practice, an equity round, a capped SAFE and a SAFE with a discount, when an accelerator invests before the seed investor and a Series A investor after it. The seed investor's best SAFE has a cap and no discount, and at a given amount an equity round gives it at least as much as a capped SAFE. The dilution rules of these contracts decide who loses from an MFN clause: the post-money SAFE does not dilute the earlier SAFE holders, while an equity round dilutes every shareholder.

The third studies most-favored-customer clauses. Cooper (1986) shows that a firm that promises to rebate its past customers if it later cuts its price has less reason to cut it, so the clause helps duopolists keep prices high. Besanko and Lyon (1993) show that a contemporaneous clause, which only covers customers served at the same time, is a weaker device, and that not adopting it can be a dominant strategy. McAfee and Schwartz (1994) show that when a supplier contracts with competing buyers, nondiscrimination clauses generally do not protect a buyer against the supplier's renegotiation with the others, which can explain why franchise contracts are uniform. In these papers the clause protects a buyer. We add to this literature by studying a clause held by an investor. Like the clause of Cooper (1986), YC's MFN SAFE gives an early signatory the better terms granted later. It converts at the cap the seed investor sets, and the seed investor raises its cap when the MFN SAFE joins its block.

The fourth studies the investors who finance a startup before VCs. Hellmann and Thiele (2015) model startups that raise angel money before VC money. The two investors are friends because each relies on the other's investment, and foes because VCs no longer need the angels at the later stage. Hellmann and Puri (2002) show that VC-backed startups professionalize faster, and Hochberg et al. (2007) show that the startups of better-networked VCs are more likely to raise a next round. On accelerators, Gonzalez-Uribe and Leatherbee (2018) find with Start-Up Chile that entrepreneurship schooling improves venture performance, while funding and coworking space alone do not. Hallen et al. (2020) find that some, but not all, accelerators speed up venture development. These papers evaluate what accelerators do, not the terms they offer. We add to this literature by modeling the accelerator's contract and its effect on the seed and Series A investors who follow.

2 Model

2.1 Agents

Suppose there is no time preference and all the agents below are risk-neutral and profit-maximizing.

A startup goes through three phases (Kerr et al., 2014). In the experimentation phase, it learns what its project is worth. In the scaling phase, it invests to grow the project. In the exit phase, the project is sold or goes public, or it fails.

Entrepreneurs have no initial budget. Each carries a project with two sunk costs. The first is a stochastic liquidity shock $l \in (0, 1)$, the cost of the experimentation phase, unknown ex ante. The second is the scaling cost I, the cost of the scaling phase, the same for all selected projects and known ex ante.

Each project yields an outcome in $\{0, 1\}$. It is worth 1, the number of shares times the price of each share, if it is sold or goes public, and 0 if it fails. Failure means that the project ends valueless and the investment is lost. The value 0 can also be read as a normalization of its salvage value.

The project's success probability $\pi \in [0, 1]$, which is also its expected outcome, is revealed once the experimentation phase is done. Before, only its distribution is known. Once the project is funded, the entrepreneur bears a private effort cost $c_e > 0$. Let f be the probability density function of π, F denote its cumulative distribution function, g be the probability density function of l, G denote its cumulative distribution function.

Three financiers take part. The accelerator and the seed investor finance the experimentation phase, and the Series A investor finances the scaling phase. The accelerator invests σ_a for a share ρ_a before l and π are known. Its terms are standard, the same for all the startups it selects, and we take them as given. The seed investor invests σ_s for a share ρ_s after the shock and before π is revealed. It is a monopolist (C1) with a limited budget β, otherwise it would fund the whole investment and Series A investors would no longer be needed. The Series A investor invests σ_A for a share ρ_A once π is revealed, in competition with other Series A investors.

2.2 Allocations and valuations

An investor who invests σ in a firm for the *share* ρ of its realized outcome values the firm at the *post-money valuation* $V = \sigma/\rho$. If the firm has N_t shares after the round and each is sold at the price p_t, then $V = N_t\, p_t$, and the *pre-money valuation* $v = N_{t-1}\, p_t = (1 - \rho)V$ is the value of the shares that existed before the round.

Seed contracts differ in the share they give. Consider two investors, A and B, who

invest one after the other, and let investor B run an equity round: it invests σ_B for the share ρ_B of the N_1 shares outstanding after its round, at the pre-money valuation v_B. Investor A's contract sets the share ρ_A it holds of the N_0 shares outstanding before this round. Investor A ends with $\rho_A(1-\rho_B)N_1$ shares, investor B with $\rho_B N_1$ and the entrepreneur with $(1-\rho_A)(1-\rho_B)N_1$. The contracts give

- with an equity round, a share ρ_A fixed when investor A signs,
- with a SAFE with a discount $\lambda \in (0,1]$, $\rho_A = \sigma_A/(\lambda v_B)$,
- with a SAFE with a cap $\bar{V}_A$ and no discount, $\rho_A = \sigma_A/\min\{\bar{V}_A, v_B\}$,
- with a SAFE with a cap $\bar{V}_A$ and a discount λ, $\rho_A = \sigma_A/\min\{\bar{V}_A, \lambda v_B\}$.

A SAFE with a cap converts at the lower of the cap and the discounted pre-money valuation of the next equity round.

An uncapped SAFE with an MFN clause takes the terms of the SAFEs the firm issues after it. Consider three investors: investor A holds an uncapped MFN SAFE, investor B a SAFE with a cap $\bar{V}_B$ and no discount, and investor C runs an equity round. Investor A's SAFE takes the cap $\bar{V}_B$, so that

$$\rho_A = \frac{\sigma_A}{\min\{\bar{V}_B, v_C\}}, \qquad \rho_B = \frac{\sigma_B}{\min\{\bar{V}_B, v_C\}},$$

and the entrepreneur ends with $\big(1-(\sigma_A+\sigma_B)/\min\{\bar{V}_B, v_C\}\big)(1-\rho_C)N_1$ shares. The two SAFEs do not dilute each other before the equity round.

Convertible notes work like SAFEs, with an interest component and a liquidation preference. Pre-money SAFEs can be diluted by other SAFEs before an equity round. Both are outside the scope of this paper.

In the model, the subscripts a, s and A refer to the accelerator, the seed investor and the Series A investor, and the Series A investor's share is $\rho_A = \sigma_A/V$. The seed investor's share ρ_s is either fixed at the seed round or set at the Series A round, as with a SAFE.

Two properties of a contract matter, and we keep them apart. A contract is *priced* when it fixes, at signing, the share the investor receives for its amount, and hence a valuation. A round is an *equity round* when it issues new shares to the investors who bring new money: it converts the SAFEs outstanding, and it dilutes every earlier shareholder in proportion to its holding. A post-money SAFE converts at the next equity round into a share of the firm measured just before the new money, and this measure includes the other SAFEs: SAFEs do not dilute one another, and each equity round dilutes them.[6] The accelerator's share ρ_a is priced: it is a fixed share of the

[6] Y Combinator's post-money SAFE forms. The company capitalization used for the conversion is "calculated as of immediately prior to the Equity Financing" and "Includes all Converting Securities", which include "other Safes". The shares issued in the equity round go "to the investors investing new money".

firm measured just before the first equity round.[7] The Series A round is an equity round.

2.3 Definitions and assumptions

Throughout, "increasing" and "decreasing" are meant in the weak sense; we write "strictly" for the strict sense.

(A1): $\mathbb{E}(\pi) < I + c_e$

$\mathbb{E}(\pi)$ is the probability of success of a project before the experiment. No investor funds the investment cost before the experiment: without l, there is no Series A round.

(A2): $\int_{I+c_e}^{1} (\pi - I - c_e)\, f(\pi)\, d\pi > \mathbb{E}(l)$

The experiment is worth running although π is unknown when it is financed, which we read as the result of the accelerator's preselection, taken as given here. Only the accelerator and the seed investor finance the experiment, and Series A investors invest once it has revealed π. Accelerators and seed investors specialize in this stage: accelerators select and train startups before handing them to venture capitalists (Gonzalez-Uribe and Leatherbee, 2018; Hallen et al., 2020), and angels finance startups before VCs do (Hellmann and Thiele, 2015).

(A3): f is positive and continuously differentiable on $(0, 1)$, and log-concave

This is a standard regularity condition satisfied by the uniform, normal, logistic, and exponential distributions, among others. It implies that the hazard rate $f/(1-F)$ is increasing and that F is log-concave (Bagnoli and Bergstrom, 2005).

(B1): $\sigma_a + \beta \leq I$

[7] Y Combinator's \$125,000 "converts into a fixed 7%" of the company, measured after the other SAFEs have converted (`https://www.ycombinator.com/deal`); the implied valuation, \$125,000/7%, is about \$1.8 million. From the Winter 2019 batch, the 7% was a post-money SAFE with a valuation cap, \$150,000 for 7% at the time; a cap gives at least this share, and more if the equity round is priced below the cap. Before, the 7% was "typically executed by buying common stock coupled with a safe" (`https://www.ycombinator.com/blog/new-standard-deal/`).

The accelerator and the seed investor together cannot fund the Series A round, so the Series A round $\sigma_A = I + l - \sigma_a - \sigma_s$ is positive whatever the shock and the seed amount. Under the reform, (B1) holds with the accelerator's cash after the reform. We do not constrain the accelerator's budget, which belongs to its preselection, and Series A investors have deep pockets.

(C1): Seed investors are monopolistic while Series A investors are competitive.

The seed investor has market power over the entrepreneur, although many investors are active at the seed stage. The seed investor acts after the shock, when the accelerator's cash does not cover the firm's need and the firm's alternative is liquidation. With a positive search cost, an investor can then make a take-it-or-leave-it offer whatever the number of investors in the market (Diamond, 1971). Series A investors arrive once the experiment has revealed π: any of them can price the project, and they compete to invest in the startups that prove good. In Crunchbase, among the first Series A rounds of Y Combinator companies from 2012 to 2024, the lead investor is new to the firm in 87% of cases, and the seed investor leads the round in 8% of firms.

2.4 Timing

The game is as follows. First, the accelerator offers its standard terms (ρ_a, σ_a), the same for all the startups it selects, before the liquidity shock. Second, a liquidity shock occurs and the seed investor proposes (ρ_s, σ_s), choosing an amount $\sigma_s \leq \beta$ and a rule defining ρ_s. A firm whose cash $\sigma_a + \sigma_s$ does not cover the shock is liquidated before the Series A round, with outcome 0. Whatever the accelerator and the seed investor pay beyond the shock stays in the firm at the Series A round. Third, the experiment reveals π to the market and enables a Series A round proposal of (σ_A, ρ_A) with $\rho_A = \sigma_A/V$, where σ_A is the part of the investment cost the firm has not yet funded, and V is the Series A valuation. Fourth, the entrepreneur bears a private effort cost c_e and payoffs are realized. Figure 1 summarizes the timing.

We take the accelerator's terms as given and study how the outcomes of its startups change with them.

The entrepreneur's outside option is zero at the accelerator stage. At the seed stage, the entrepreneur can turn down the offer: the firm then continues to the Series A round if $l \leq \sigma_a$ and is liquidated otherwise.

3 Benchmark: equity rounds only

We first solve the game when the seed investor and the Series A investor raise equity rounds and the accelerator holds a fixed share. Each equity round issues new shares,

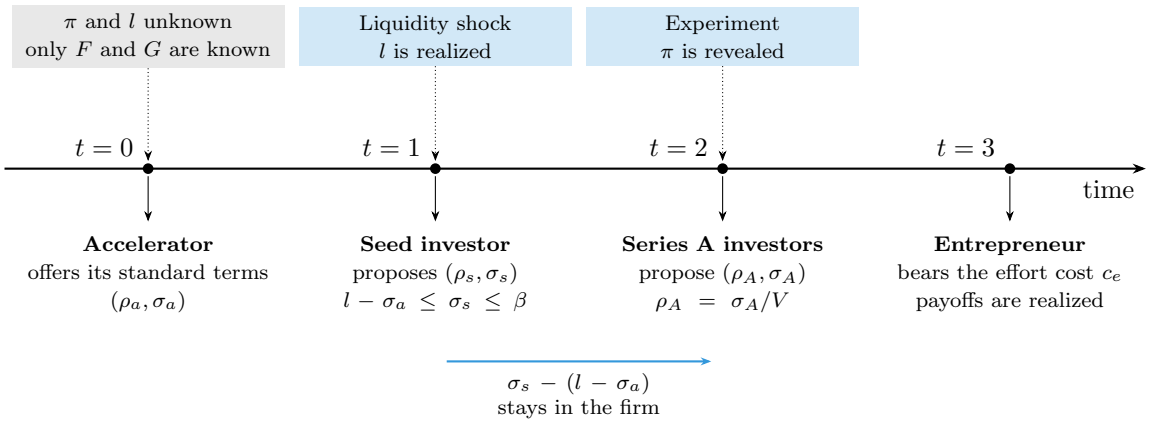


Figure 1: Timing of the game.

so it dilutes every earlier shareholder in proportion to its holding. The accelerator's share is the same whether it is signed as common stock or as a post-money SAFE: in both cases it is a share of the firm measured before the first equity round, and this round dilutes it. The seed investor covers the firm's need exactly, and more accelerator cash lowers no firm's probability of reaching Series A. We solve by backward induction.

3.1 Entrepreneur

The seed round dilutes the accelerator and the entrepreneur by the factor $1 - \rho_s$, and the Series A round dilutes all three by the factor $1 - \rho_A$. The utility u_e of an entrepreneur carrying a project π writes:

$$u_e = (1 - \rho_a)(1 - \rho_s)(1 - \rho_A)\pi - c_e \tag{1}$$

Then, the entrepreneur's incentive compatibility constraint writes:

$$\pi \geq \frac{c_e}{(1 - \rho_a)(1 - \rho_s)(1 - \rho_A)} \tag{IC_e}$$

c_e can also be read as the private benefits the entrepreneur would get from not working once funded. To make the project work, the entrepreneur gives them up.

3.2 Series A investor

For each financed project π, the Series A investor sets a valuation $V(\pi)$ that satisfies its participation constraint:

$$\frac{\sigma_A}{V(\pi)}\pi - \sigma_A \geq 0 \qquad (IR_A)$$

Series A investors compete, so the entrepreneur obtains the highest valuation that satisfies this constraint:

$$V^*(\pi) = \pi. \tag{2}$$

With $\rho_A = \sigma_A/\pi$, (IC_e) reads $(1-\rho_a)(1-\rho_s)(\pi - \sigma_A) \geq c_e$, so the Series A investor finances the projects above

$$\pi_{\min} = \sigma_A + \frac{c_e}{(1-\rho_a)(1-\rho_s)} \tag{3}$$

The Series A round raises the threshold one for one, because the Series A investor's share dilutes every earlier shareholder. The shares of the accelerator and of the seed investor raise it through the effort cost alone.

3.3 Seed investor

The seed investor invests when the shock must be covered, before the market reveals π. Its share σ_s/V_s is diluted by the Series A round, so on a financed project it receives $\frac{\sigma_s}{V_s}\big(1 - \frac{\sigma_A}{\pi}\big)\pi = \frac{\sigma_s}{V_s}(\pi - \sigma_A)$. It solves

$$\begin{aligned}\max_{\sigma_s, V_s} \Pi_s &= \max_{\sigma_s, V_s} \int_{\pi_{\min}}^{1} \Big(\frac{\sigma_s}{V_s}(\pi - \sigma_A) - \sigma_s\Big) f(\pi)\, d\pi \;-\; \int_0^{\pi_{\min}} \sigma_s\, f(\pi)\, d\pi \\ &= \max_{\sigma_s, V_s} \sigma_s \left(\frac{N(\pi_{\min}; \sigma_A)}{V_s} - 1 \right).\end{aligned}$$

with:

$$M(x) \;:=\; \int_x^1 \pi\, f(\pi)\, d\pi, \qquad N(x;\sigma_A) \;:=\; \int_x^1 (\pi - \sigma_A)\, f(\pi)\, d\pi$$

so that $N(x;\sigma_A) = M(x) - \sigma_A\big(1 - F(x)\big)$.

$M(x)$ is the expected outcome contributed by projects above x, and $N(x;\sigma_A)$ is that outcome net of the Series A round.

The seed investor knows that some projects will not reach the Series A round, but not which ones. A lower seed valuation gives it a larger share, but it also dilutes the entrepreneur and raises the Series A threshold: the seed investor must leave the entrepreneur enough of the firm to exert effort. This is the financing risk of Nanda and Rhodes-Kropf (2017).

When the shock exceeds the accelerator's cash, $l > \sigma_a$, the firm needs $l - \sigma_a$ more to run the experiment.

Lemma 1. *Under each seed contract of Sections 3 to 6, a seed investor who invests less than the need, $\sigma_s \in (0,\, l - \sigma_a)$, earns $-\sigma_s < 0$, and no project reaches Series A. The seed investor therefore invests nothing or at least the need, $\sigma_s \geq l - \sigma_a$.*

→ Proof: Appendix, page 38

A participating seed investor therefore advances $\sigma_s = l - \sigma_a + \Delta$ with $\Delta \geq 0$. The amount Δ is not needed to run the experiment: it stays in the firm and prefinances the Series A round, which falls to $\sigma_A = I - \Delta$.

Fix the amount σ_s and the Series A round σ_A. Choosing the valuation V_s is choosing the share $\rho_s = \sigma_s / V_s$, and the share sets the threshold through (3). Inverting (3),

$$\rho_s \;=\; 1 - \frac{c_e}{(1-\rho_a)(\pi_{\min} - \sigma_A)},$$

so the seed investor chooses the threshold. With the threshold x, its claims are worth the first term below, and it earns

$$\Big(1 - \frac{c_e}{(1-\rho_a)(x - \sigma_A)}\Big) N(x; \sigma_A) - \sigma_s. \tag{4}$$

The share is positive only if $x > \sigma_A + c_e/(1-\rho_a)$, and a project is financed only if $x < 1$. Such a threshold exists if and only if

$$\sigma_A \;<\; 1 - \frac{c_e}{1-\rho_a}. \tag{5}$$

Otherwise even the project that succeeds for sure is worth $1 - \sigma_A \leq c_e/(1-\rho_a)$ once the Series A investor has its share, and the entrepreneur, who holds at most $1 - \rho_a$ of it, cannot cover the effort cost. No project reaches Series A and the seed investor's claims are worth nothing.

When (5) holds, the derivative of (4) in x, with $\partial N(x;\sigma_A)/\partial x = -(x - \sigma_A) f(x)$, vanishes where

$$(1-\rho_a)\, f(x)\, (x - \sigma_A)^3 \;=\; c_e \Big(N(x;\sigma_A) + f(x)\, (x - \sigma_A)^2 \Big). \tag{6}$$

Proposition 1. *Under* (5)*, equation* (6) *has a unique solution $\pi(\sigma_A)$ in $(\sigma_A, 1)$. The seed investor sets the threshold $\pi(\sigma_A)$, with the valuation*

$$V_s(\sigma_s, \sigma_A) = \frac{\sigma_s}{1 - \dfrac{c_e}{(1-\rho_a)\big(\pi(\sigma_A) - \sigma_A\big)}}.$$

→ Proof: Appendix, page 38

The threshold depends on the accelerator's share, the distribution of projects, the effort cost and the Series A round. The seed valuation depends in addition on the amount invested. Write

$$W(\sigma_A) \;:=\; \Big(1 - \frac{c_e}{(1-\rho_a)\big(\pi(\sigma_A) - \sigma_A\big)}\Big) N\big(\pi(\sigma_A); \sigma_A\big)$$

for the value of the seed investor's claims at this threshold under (5), and $W(\sigma_A) := 0$ otherwise. A firm whose seed investor covers the need exactly faces the Series A round I, and a firm without a seed round faces $I + l - \sigma_a$.

Corollary 1. *$\pi(\cdot)$ is strictly increasing, and $\pi(\sigma_A) > \sigma_A + c_e/(1-\rho_a)$.*

→ Proof: Appendix, page 40

A dollar the seed investor invests beyond the need stays in the firm and lowers the Series A round by a dollar. It adds a dollar to the net outcome of every financed project, of which the seed investor holds its share, and it lets the seed investor take a larger share at the same threshold. These gains never cover the dollar, which is lost on the projects that the Series A investor does not finance. Write Δ^* for the amount the seed investor invests beyond the need, the largest when it is indifferent, $\sigma_s^* := l - \sigma_a + \Delta^*$ for its investment and $\sigma_A^* := I - \Delta^*$ for the Series A round it leaves.

Proposition 2. *A participating seed investor covers the need exactly, $\sigma_s^* = l - \sigma_a$: the Series A round is I, and the Series A investor finances the projects above $\pi(I)$.*

→ Proof: Appendix, page 40

Investing beyond the need lowers the share of the firm the entrepreneur must keep to exert effort, which is the entrepreneur's information rent, and lets the seed investor take a larger share. The cost is the cash lost on the projects below the threshold. Under (A1), financing every project would destroy value, and with the log-concavity of (A3) the cost always exceeds the gain.

3.4 Liquidity shock

The liquidity shock may force the entrepreneur to raise funds before the Series A round. When $l \le \sigma_a$ the firm can reach the Series A round without a seed round: it keeps $\sigma_a - l$ and the Series A round is $\sigma_A = I + l - \sigma_a$. Write $\pi_0(l) := I + l - \sigma_a + c_e/(1-\rho_a)$ for the Series A threshold of such a firm, that is (3) with $\rho_s = 0$ and $\sigma_A = I + l - \sigma_a$.

Lemma 2. *A firm with $l \le \sigma_a$ has no seed round: the entrepreneur turns down every seed offer that gives the seed investor a non-negative profit. This holds under each seed contract of Sections 3 to 6.*

→ Proof: Appendix, page 42

The seed investor's share dilutes the entrepreneur, and the cash it advances is lost on the projects below the Series A threshold.

When $l > \sigma_a$, turning the seed offer down liquidates the firm, so the entrepreneur accepts. The seed investor covers the need $l - \sigma_a$ exactly (Proposition 2), leaves the Series A round I and earns $W(I) - (l - \sigma_a)$. It takes part if this profit is non-negative and the need fits in its budget, that is if

$$l - \sigma_a \;\leq\; \min\{\beta,\, W(I)\}.$$

Write $\bar{l} := \sigma_a + \min\{\beta, W(I)\}$ for the largest shock with a seed round.

Proposition 3. *The liquidity shock sorts projects into three regimes:*

(N) No seed round, $l \leq \sigma_a$*: the Series A investor finances the projects* $\pi \geq \pi_0(l)$.

(P) Seed round, $\sigma_a < l \leq \bar{l}$*: the Series A investor finances the projects* $\pi \geq \pi(\sigma_A^*)$.

(L) Liquidation, $l > \bar{l}$*: the firm is liquidated before Series A.*

→ Proof: Appendix, page 43

The accelerator covers the smallest shocks. Larger ones need a seed round, and beyond $\bar{l}$ the seed investor no longer participates and the firm is liquidated. The seed investor participates up to the need at which its budget or the value of its claims runs out. Neither depends on σ_a, so $\bar{l} - \sigma_a$ does not either.

Figure 2 shows the three regimes.

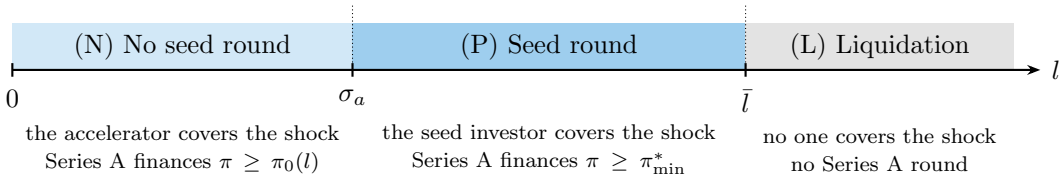


Figure 2: The three regimes of Proposition 3.

3.5 Accelerator: increasing investment amount

The accelerator's amount rises from σ_a to $\sigma_a' > \sigma_a$. Since $\bar{l} - \sigma_a$ does not depend on σ_a (Proposition 3), the largest shock with a seed round moves from $\bar{l}$ to $\bar{l}' := \bar{l} + \sigma_a' - \sigma_a$. The three regimes are the events $l \leq \sigma_a$, $\sigma_a < l \leq \bar{l}$ and $l > \bar{l}$, so their probabilities move with the two bounds. The probability of a seed round goes from $G(\bar{l}) - G(\sigma_a)$ to $G(\bar{l}') - G(\sigma_a')$.

Proposition 4. *The increase:*

(i) raises the probability that a firm needs no seed round from $G(\sigma_a)$ *to* $G(\sigma_a')$*;*

(ii) raises the probability of a seed round if and only if $G(\bar{l}') - G(\bar{l}) > G(\sigma_a') - G(\sigma_a)$*;*

(iii) lowers the probability that a firm is liquidated from $1 - G(\bar{l})$ *to* $1 - G(\bar{l}')$*.*

→ Proof: Appendix, page 44

The two bounds σ_a and $\bar{l}$ move by the same amount. Firms leave the seed round regime at the bottom and enter it at the top, so seed rounds become more frequent only if more firms enter at the top than leave at the bottom.

With equity rounds, more accelerator cash raises the probability that no seed round is needed and lowers the probability of liquidation. It does not necessarily raise the probability of a seed round. If large shocks are the most likely, many firms that were liquidated now raise a seed round, and seed rounds become more frequent. If small shocks are the most likely, many firms that needed a seed round no longer do, and seed rounds become less frequent. Its effect on the probability of reaching Series A depends on the amount the seed investor invests: a firm that no longer needs a seed round also loses the cash its seed investor invested beyond its need. Since the seed investor covers the need exactly (Proposition 2), no firm loses such cash.

After the increase, a firm with $l \leq \sigma_a'$ needs no seed round and has the threshold

$$\pi_0'(l) \;:=\; I + l - \sigma_a' + \frac{c_e}{1 - \rho_a}.$$

This threshold is below the one the firm had before: $\pi_0(l)$ if $l \leq \sigma_a$, and $\pi(I)$ if it had a seed round, since $\pi_0'(l) \leq I + c_e/(1 - \rho_a) < \pi(I)$ (Corollary 1). The firm reaches Series A with a positive probability if and only if $\pi_0'(l) < 1$. A firm with $l > \sigma_a'$ keeps the threshold $\pi(I)$ if it had a seed round, and the firms with $\bar{l} < l \leq \bar{l}'$ obtain one.

Proposition 5. *The increase has the following effects:*

(i) no firm's probability of reaching Series A falls;

(ii) the probability of a firm with $l \leq \sigma_a'$ *rises if and only if* $\pi_0'(l) < 1$*, and the probability of a firm with* $\max\{\sigma_a', \bar{l}\} < l \leq \bar{l}'$ *rises;*

(iii) the probability of every other firm does not change.

→ Proof: Appendix, page 44

Integrating over shocks, the probability of reaching Series A does not fall, and it rises when G puts a positive probability on the firms of part (ii).

4 The MFN SAFE with a seed equity round

The accelerator now holds its share ρ_a^0 as a post-money SAFE and, besides it, an uncapped SAFE of amount m with an MFN clause: the SAFE takes the terms of the SAFEs the firm issues after it, and it converts at the price of the next equity round if there is none.[8] Read $\pi(\cdot)$, W and (5) of Section 3 with ρ_a^0 in place of ρ_a.

The seed investor raises an equity round: it invests σ_s for the share σ_s/V_s of the firm after the seed round. The seed round is the first equity round, so it converts the accelerator's SAFEs. The MFN SAFE is uncapped and converts at the seed price, for the share m/V_s. The accelerator's post-money SAFE gives it ρ_a^0 of the firm before the seed investor's cash, so the seed round dilutes it by the factor $1-\sigma_s/V_s$, and the conversion of the MFN SAFE does not. Before the Series A round the entrepreneur holds

$$e \;=\; (1-\rho_a^0)\Big(1-\frac{\sigma_s}{V_s}\Big)-\frac{m}{V_s},$$

and the Series A round dilutes every shareholder. With $m=0$ this is the benchmark of Section 3.

Without a seed round, no equity round precedes the Series A round. The MFN SAFE then converts at the Series A price, its claims are worth m on every financed project, and a firm with $l \le \sigma_a$ has the threshold

$$\pi_0^m(l) \;:=\; I + l - \sigma_a + \frac{c_e+m}{1-\rho_a^0}.$$

4.1 The seed investor's price

(IC_e) and $V^*(\pi)=\pi$ give $e\,(\pi-\sigma_A)\ge c_e$, that is

$$\pi_{\min} \;=\; \sigma_A + \frac{c_e}{e}.$$

As in the benchmark, choosing the valuation is choosing the threshold x, through $e=c_e/(x-\sigma_A)$. Since $e=(1-\rho_a^0)-\big((1-\rho_a^0)\sigma_s+m\big)/V_s$, the seed investor holds

$$\frac{\sigma_s}{V_s} \;=\; \frac{(1-\rho_a^0)\sigma_s}{(1-\rho_a^0)\sigma_s+m}\Big(1-\frac{c_e}{(1-\rho_a^0)(x-\sigma_A)}\Big)$$

of the firm before the Series A round. Its claims are worth the fraction $(1-\rho_a^0)\sigma_s/\big((1-\rho_a^0)\sigma_s+m\big)$ of the claims of a benchmark seed investor at the same threshold, the first term of (4). The fraction does not depend on x, so the seed investor sets the threshold of the benchmark.

[8] In Y Combinator's forms, the MFN clause covers the "Subsequent Convertible Securities", that is "other Safes, convertible debt instruments and other convertible securities", and not the shares of an equity round.

Proposition 6. *Under* (5)*, the seed investor sets the threshold* $\pi(\sigma_A)$ *and proposes*

$$V_s^* \;=\; \frac{\big((1-\rho_a^0)\sigma_s + m\big)\big(\pi(\sigma_A) - \sigma_A\big)}{(1-\rho_a^0)\big(\pi(\sigma_A) - \sigma_A\big) - c_e}.$$

It earns

$$\frac{(1-\rho_a^0)\sigma_s}{(1-\rho_a^0)\sigma_s + m}\, W(\sigma_A) - \sigma_s.$$

→ Proof: Appendix, page 45

The entrepreneur keeps $c_e/\big(\pi(\sigma_A) - \sigma_A\big)$ of the firm before the Series A round, as in the benchmark. The MFN SAFE takes part of what the seed investor would hold.

4.2 The amount

Without the MFN SAFE, the seed investor covers the need exactly (Proposition 2). With it, the fraction $(1-\rho_a^0)\sigma_s/\big((1-\rho_a^0)\sigma_s + m\big)$ of the claims the seed investor holds rises with its amount. At the need $b = l - \sigma_a$, the amount $\sigma_s \geq b$ leaves the Series A round $I + b - \sigma_s$, and by Proposition 6 the seed investor earns

$$\frac{(1-\rho_a^0)\sigma_s}{(1-\rho_a^0)\sigma_s + m}\, W(I + b - \sigma_s) - \sigma_s.$$

W is decreasing (proof of Proposition 2), so at exact funding, $\sigma_s = b$, the right derivative of this profit is at least

$$\frac{(1-\rho_a^0)\, m}{\big((1-\rho_a^0)b + m\big)^2}\, W(I) - 1,$$

which tends to $(1-\rho_a^0)W(I)/m - 1$ as the need goes to zero. It is positive when

$$0 \;<\; m \;<\; (1-\rho_a^0)\, W(I), \tag{7}$$

that is when the MFN SAFE exists and the claims of the seed investor and the MFN SAFE are worth more than m at exact funding.

Proposition 7. *Under* (7)*, there is* $b_0 > 0$ *such that the seed investor invests beyond the need at every need* $l - \sigma_a \in (0, b_0)$*.*

→ Proof: Appendix, page 46

4.3 Participation and regimes

Write σ_s^* for the amount the seed investor invests, the largest when it is indifferent, and $\sigma_A^* := I + l - \sigma_a - \sigma_s^*$.

Proposition 8. *There is $\bar{l}$, with $\bar{l} - \sigma_a \in [0, \beta]$ depending on f, I, c_e, ρ_a^0, β and m but not on σ_a, and weakly decreasing in m, such that the liquidity shock sorts projects into three regimes:*

(N) No seed round, *$l \leq \sigma_a$: the Series A investor finances the projects $\pi \geq \pi_0^m(l)$.*

(P) Seed round, *$\sigma_a < l \leq \bar{l}$: the Series A investor finances the projects $\pi \geq \pi(\sigma_A^*)$.*

(L) Liquidation, *$l > \bar{l}$: the firm is liquidated before Series A.*

→ Proof: Appendix, page 46

At a given cash σ_a, the MFN SAFE raises the threshold of the firms without a seed round and narrows the set of shocks with a seed round.

4.4 The reform

Under Y Combinator's reform the accelerator pays m more in cash, and this m is the MFN SAFE: the cash goes from σ_a^0 to $\sigma_a^0 + m$, and the share ρ_a^0 does not change. Before the reform the accelerator holds no MFN SAFE, and the seed equity round is the benchmark of Section 3 with ρ_a^0 in place of ρ_a: a participating seed investor covers the need exactly and leaves the threshold $\pi(I)$ (Proposition 2). We compare the probability of reaching Series A before and after the reform, shock by shock. Write $x := l - \sigma_a^0$ for the need the former cash leaves and $\pi_0 := I + x + c_e/(1 - \rho_a^0)$ for the threshold of a firm without a seed round before the reform. The reform sorts the shocks into three groups.

The former cash covers the shock, $x \leq 0$. The firm has no seed round before or after the reform (Lemma 2). After the reform the Series A round is $I + x - m$ and the MFN SAFE converts at the Series A price, so the threshold $\pi_0^m(l)$ is

$$I + x - m + \frac{c_e + m}{1 - \rho_a^0} \;=\; \pi_0 + \frac{\rho_a^0\, m}{1 - \rho_a^0}.$$

The MFN SAFE does not dilute the accelerator's share, while the Series A round it replaces did.

The new cash covers the shock, $0 < x \leq m$. After the reform the firm has no seed round, and its threshold is $\pi_0 + \rho_a^0 m/(1 - \rho_a^0)$ as well. Before it, the firm had a seed round with the threshold $\pi(I)$ if $x \leq \min\{\beta, W(I)\}$ (Proposition 3), and it was liquidated otherwise.

The shock exceeds the new cash, $x > m$. The firm needs a seed round before and after the reform. Write S for the amount placed beyond the former cash, σ_s before the reform and $m+\sigma_s$ after it, so that the Series A round is $I+x-S$ in both cases. Before the reform the seed investor earns $W(I+x-S)-S$ (Proposition 6 with $m=0$). After it, it invests $\sigma_s = S-m$ and earns

$$\frac{(1-\rho_a^0)(S-m)}{(1-\rho_a^0)(S-m)+m}\Big(W(I+x-S)-S-\frac{\rho_a^0\, m}{1-\rho_a^0}\Big).$$

The fraction is positive, so the seed investor participates after the reform if and only if, at some amount $S \in [x, \beta+m]$, its profit before the reform is at least $\rho_a^0 m/(1-\rho_a^0)$. Before the reform the best amount is the need, whatever the budget (proof of Proposition 2), and the profit there is $W(I)-x$. The seed investor therefore participates after the reform if and only if

$$x \;\le\; \min\Big\{\beta+m,\; W(I)-\frac{\rho_a^0\, m}{1-\rho_a^0}\Big\}. \tag{8}$$

It then places $S \ge x$, and the threshold $\pi(I+x-S)$ is at most $\pi(I)$ (Corollary 1). Otherwise the shock exceeds the new cash and the firm is liquidated. Before the reform the firm had a seed round if and only if $x \le \min\{\beta, W(I)\}$.

Proposition 9. *The reform changes the probability of reaching Series A as follows.*

(i) For $x \le 0$, it falls if $\pi_0 < 1$.

(ii) For $0 < x \le m$, it falls if and only if the firm had a seed round and $x > \pi(I) - I - (c_e + \rho_a^0 m)/(1-\rho_a^0)$.

(iii) For $x > m$, it falls to zero for the firms with $W(I) - \rho_a^0 m/(1-\rho_a^0) < x \le \min\{\beta, W(I)\}$, which lose their seed round, and it does not fall for the others. The firms with $\beta < x \le \min\{\beta+m,\, W(I)-\rho_a^0 m/(1-\rho_a^0)\}$ obtain a seed round.

→ Proof: Appendix, page 47

With a seed equity round the reform can liquidate a firm which had a seed round: the MFN SAFE takes part of the seed investor's claims, and a firm whose need comes within $\rho_a^0 m/(1-\rho_a^0)$ of the value $W(I)$ of these claims loses its seed round. Figure 3 shows the groups. With a small budget the seed investor's budget stopped the firms beyond $\sigma_a^0+\beta$, and the reform gives them a seed round; with a large one the firms just below $\bar{l}$ lose theirs.

5 The MFN SAFE with a capped seed SAFE

The accelerator holds its post-money SAFE ρ_a^0 and the MFN SAFE m of Section 4. The seed investor now holds a SAFE with a cap V_s and a discount: at the Series

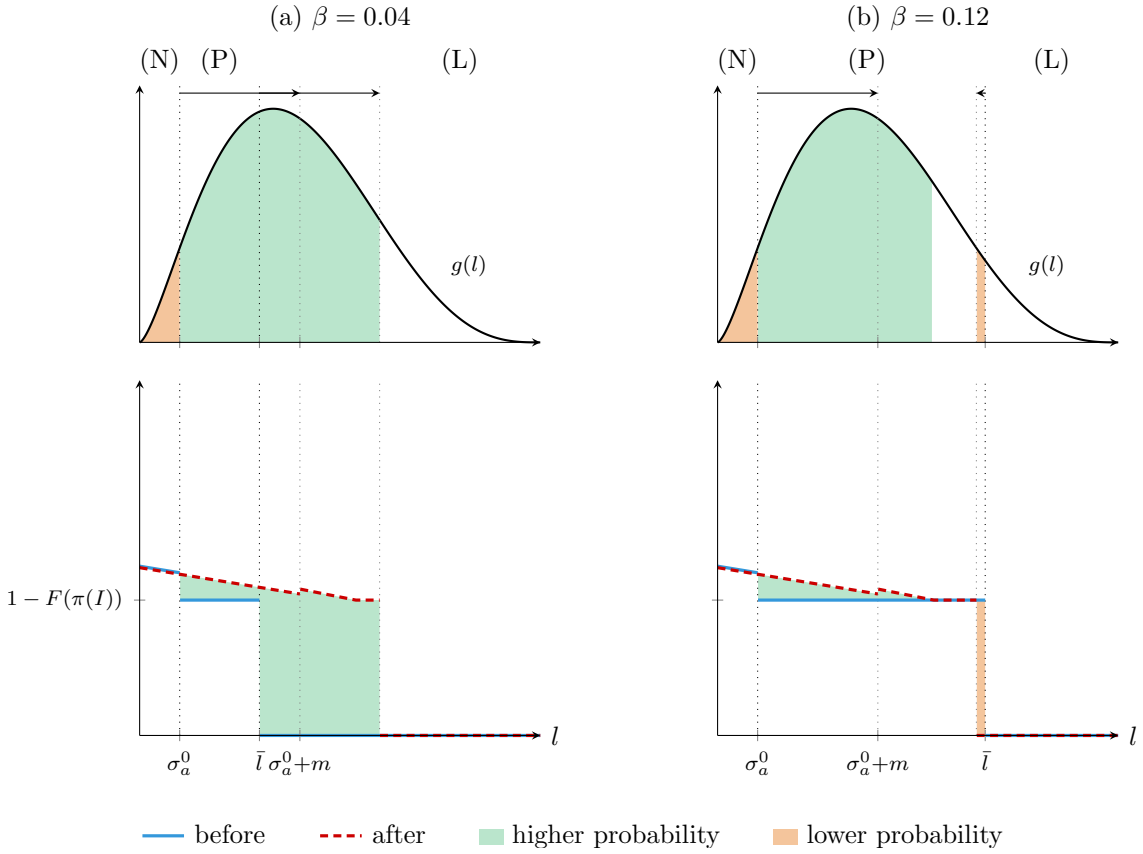


Figure 3: The probability of reaching Series A before and after the reform, by liquidity shock, when the seed investor raises an equity round (Proposition 9). The project's probability of success is uniform on $[0, 1]$, $I = 0.5$, $c_e = 0.005$, $\rho_a^0 = 0.07$, $\sigma_a^0 = 0.02$ and $m = 0.06$; the shock l follows a Beta$(2.5, 4)$ distribution on $[0, 0.2]$, whose density g is in the top panels.

A round, the SAFE converts at the lower of the cap and the fraction $\lambda \in (0,1]$ of the Series A pre-money valuation. The MFN clause gives the accelerator's SAFE the same cap and the same discount.[9] In this section the discount is given, and Section 6 studies how it is set.

All SAFEs are post-money SAFEs. The accelerator's share ρ_a^0 and the shares of the SAFEs are set in the firm before the Series A round, so the SAFEs do not dilute the accelerator, and the Series A round dilutes all of them. The SAFE block is $m + \sigma_s$.

5.1 Entrepreneur and Series A investor

As in the benchmark, Series A investors compete and $V^*(\pi) = \pi$, so the pre-money valuation of the project π at the Series A round is $\pi - \sigma_A$. At the cap the SAFE block takes the share $(m+\sigma_s)/V_s$ of the firm before the Series A round. At the discount it receives shares worth $(m+\sigma_s)/\lambda$, that is the share $(m+\sigma_s)/\big(\lambda(\pi-\sigma_A)\big)$. It takes the larger share, so it converts at $\min\{V_s, \lambda(\pi-\sigma_A)\}$, and the entrepreneur's incentive constraint reads

$$\Big(1-\rho_a^0-\frac{m+\sigma_s}{\min\{V_s,\lambda(\pi-\sigma_A)\}}\Big)(\pi-\sigma_A) \;\geq\; c_e.$$

Case 1: Project converts at cap ($V_s \leq \lambda(\pi-\sigma_A)$).

$$\Big(1-\rho_a^0-\frac{m+\sigma_s}{V_s}\Big)(\pi-\sigma_A) \;\geq\; c_e, \qquad \pi \;\geq\; \sigma_A + \frac{c_e}{1-\rho_a^0-(m+\sigma_s)/V_s} \;=\; \pi^c_{\min}$$

Case 2: Project converts at discount ($V_s \geq \lambda(\pi-\sigma_A)$).

$$(1-\rho_a^0)(\pi-\sigma_A) - \frac{m+\sigma_s}{\lambda} \;\geq\; c_e, \qquad \pi \;\geq\; \sigma_A + \frac{c_e+(m+\sigma_s)/\lambda}{1-\rho_a^0} \;=\; \pi^d_{\min}$$

With $\sigma_A = I+l-\sigma_a-\sigma_s$, $\pi^d_{\min} = I+l-\sigma_a+\big(c_e+m/\lambda\big)/(1-\rho_a^0)+\big(1/(\lambda(1-\rho_a^0))-1\big)\sigma_s$ rises with the amount, since $\lambda(1-\rho_a^0) < 1$: in Case 2 the block holds claims worth $(m+\sigma_s)/\lambda$ that do not dilute the accelerator, so a dollar moved from the Series A round to the block lowers the entrepreneur's payoff on every financed project by $1/\lambda - (1-\rho_a^0)$.

[9] In Y Combinator's forms, a SAFE with a valuation cap converts into "the greater of" the shares bought at the price of the equity round and the shares bought at the cap.

5.2 Seed investor

The seed investor holds the fraction $\sigma_s/(m+\sigma_s)$ of the SAFE block and earns

$$\Pi_s \;=\; \sigma_s\Big(\int_{\text{financed}} \frac{\pi-\sigma_A}{\min\{V_s,\lambda(\pi-\sigma_A)\}}\, f(\pi)\, d\pi - 1\Big).$$

The cap sets the threshold. If every financed project converts at the cap, the threshold x satisfies the constraint of Case 1 with equality, so the SAFE block holds $1-\rho_a^0-c_e/(x-\sigma_A)$ of the firm before the Series A round, and its claims are worth

$$\Big(1-\rho_a^0-\frac{c_e}{x-\sigma_A}\Big)N(x;\sigma_A),$$

$1-\rho_a^0$ times the first term of (4), the claims of a benchmark seed investor at the same threshold. Raising the cap lowers this threshold down to $\pi^d_{\min}$, where the marginal project converts at the discount. A higher cap lets the marginal projects convert at the discount without lowering the threshold further. Choosing the cap is therefore choosing a threshold $x \geq \pi^d_{\min}$. By Proposition 1, the value of the claims increases up to $\pi(\sigma_A)$ and decreases beyond it, which gives two cases:

- Case 1, $\pi(\sigma_A) \geq \pi^d_{\min}$: the seed investor can set the threshold $\pi(\sigma_A)$ of a seed equity round;
- Case 2, $\pi(\sigma_A) < \pi^d_{\min}$: this threshold is out of reach, and the seed investor sets the lowest threshold it can reach, $\pi^d_{\min}$.

Proposition 10. *Under* (5)*, the seed investor sets the threshold* x^**, the larger of* $\pi(\sigma_A)$ *and* $\pi^d_{\min}$*, with the cap*

$$V_s^* \;=\; \frac{(m+\sigma_s)(x^*-\sigma_A)}{(1-\rho_a^0)(x^*-\sigma_A)-c_e},$$

which equals $(\lambda c_e+m+\sigma_s)/(1-\rho_a^0)$ *in Case 2. Every financed project converts at the cap.*

→ Proof: Appendix, page 48

In Case 1 the entrepreneur keeps $c_e/\big(\pi(\sigma_A)-\sigma_A\big)$ of the firm before the Series A round, as with a seed equity round.

5.3 The amount

For each need $b = l-\sigma_a \in (0,\beta]$, the seed investor chooses its amount in $[b,\beta]$ (Lemma 1). It holds only the fraction $\sigma_s/(m+\sigma_s)$ of the block, and a dollar it adds raises that fraction. In Case 1 it earns

$$\frac{\sigma_s}{m+\sigma_s}\,(1-\rho_a^0)\,W(I+b-\sigma_s)-\sigma_s,$$

and the computation of Section 4 gives, at exact funding, a right derivative that tends to $(1-\rho_a^0)W(I)/m - 1$ as the need goes to zero. At exact funding, $\pi^d_{\min} = I + \big(c_e + (m+b)/\lambda\big)/(1-\rho_a^0)$, which tends to $I + (c_e + m/\lambda)/(1-\rho_a^0)$, so Case 1 holds at exact funding for small needs when

$$\pi(I) \;>\; I + \frac{c_e + m/\lambda}{1-\rho_a^0}. \tag{9}$$

Proposition 11. *(i) If $m = 0$, the seed investor covers the need exactly: $\sigma_s^* = l - \sigma_a$.*

(ii) Under (7) and (9), there is $b_0 > 0$ such that the seed investor invests beyond the need, $\sigma_s^* > l - \sigma_a$, at every need $l - \sigma_a \in (0, b_0)$.

→ Proof: Appendix, page 49

5.4 Participation and regimes

Write σ_s^* for the amount the seed investor invests, the largest when it is indifferent, and $\sigma_A^* := I + l - \sigma_a - \sigma_s^*$.

Proposition 12. *There is $\bar{l}$, with $\bar{l} - \sigma_a \in [0, \beta]$ depending on f, I, c_e, ρ_a^0, β, m and λ but not on σ_a, and weakly decreasing in m, such that the liquidity shock sorts projects into three regimes:*

(N) No seed round, *$l \le \sigma_a$: the Series A investor finances the projects $\pi \ge \pi_0^m(l)$.*

(P) Seed round, *$\sigma_a < l \le \bar{l}$: the Series A investor finances the projects $\pi \ge \max\big\{\pi(\sigma_A^*),\, \pi^d_{\min}\big\}$, evaluated at the seed investor's amount.*

(L) Liquidation, *$l > \bar{l}$: the firm is liquidated before Series A.*

→ Proof: Appendix, page 50

At a given cash σ_a, the MFN SAFE narrows the set of shocks with a seed round: it raises the threshold of Case 2 and gives part of the block to the accelerator.

5.5 The reform

As in Section 4.4, the reform raises the accelerator's cash from σ_a^0 to $\sigma_a^0 + m$ and gives it the MFN SAFE m, and we keep x, π_0 and the three groups of shocks. Before the reform $m = 0$, a participating seed investor covers the need exactly (Proposition 11(i)), the Series A round is I, and by Proposition 10 with $\sigma_s = x$ the threshold is

$$\pi^b(x) \;:=\; \max\Big\{\pi(I),\; I + \frac{c_e + x/\lambda}{1-\rho_a^0}\Big\}.$$

Case 2 holds before the reform if and only if the second term is the larger, that is

$$x \;>\; \lambda\Big((1-\rho_a^0)\big(\pi(I)-I\big)-c_e\Big), \tag{10}$$

a positive bound by Corollary 1.

The cash covers the shock, $x \le m$. As with a seed equity round, the firm has no seed round after the reform, and its threshold is $\pi_0+\rho_a^0 m/(1-\rho_a^0)$. Before it, the threshold was π_0 if $x \le 0$, and $\pi^b(x)$ or liquidation if $0 < x \le m$.

The shock exceeds the new cash, $x > m$. Write again S for the amount placed beyond the former cash, σ_s before the reform and $m+\sigma_s$ after it. The SAFE block is S in both cases and the Series A round is $I+x-S$, so by Proposition 10 the threshold is

$$T(S) \;:=\; \max\Big\{\pi(I+x-S),\; \pi_0+\Big(\frac{1}{\lambda(1-\rho_a^0)}-1\Big)S\Big\} \tag{11}$$

before and after the reform, and the value of the block's claims at the best cap depends on S only. The reform does not change the map from the amount to the threshold, only the amount. The first term is strictly decreasing in S (Corollary 1) and the second strictly increasing, and Case 2 holds where the second is the larger: T decreases in Case 1 and increases in Case 2.

Before the reform the seed investor holds the whole block and earns the value of its claims minus S. After it, it holds the fraction $(S-m)/S$ of the block, which rises with S, and earns $(S-m)/S$ times the same difference. A seed investor which participated before the reform, at $S=x$, therefore earns a non-negative profit at $S=x$ after it: a firm with $x > m$ which had a seed round keeps it. The seed investor's new need is $x-m$, so it places $S \ge x$, and the threshold moves from $T(x)=\pi^b(x)$ to $T(S)$. If Case 1 holds at S, it holds on $[x,S]$, and the threshold does not rise. Under (10), Case 2 holds at x, hence on $[x,S]$, and the threshold does not fall; it rises if $S > x$, that is if the seed investor invests beyond the need after the reform.

Proposition 13. *The reform changes the probability of reaching Series A as follows.*

(i) For $x \le 0$, it falls if $\pi_0 < 1$.

(ii) For $0 < x \le m$, it falls if and only if the firm had a seed round and $\pi^b(x) < \pi_0+\rho_a^0 m/(1-\rho_a^0)$.

(iii) For $x > m$, a firm which had a seed round keeps it. Its probability does not fall if Case 1 holds at the amount it receives after the reform. Under (10), its probability does not rise, and it falls if the seed investor invests beyond the need after the reform. The probability of a firm which was liquidated does not fall.

→ Proof: Appendix, page 51

Table 1 collects the groups.

Shock	Before the reform	Probability of reaching Series A
(i) $x \leq 0$	No seed round	$-$
(ii) $0 < x \leq m$	Liquidated	$+$
	Seed round	$+$ or $-$
(iii) $x > m$	Liquidated	0 or $+$
	Seed round, Case 1 after the reform	$+$
	Seed round, Case 2, (10)	$-$
	Seed round, Case 1 before and Case 2 after	$+$ or $-$

Table 1: Effect of the reform on the probability of reaching Series A, by shock. The signs $+$ and $-$ are weak.

A firm with a small shock loses: the m added to its cash lowers the Series A round, which dilutes the accelerator, and the m added to its claims does not dilute the accelerator. A firm whose shock is just above the former cash exchanges a seed round for the accelerator's additional cash. A firm with a larger shock keeps its seed round and receives a larger amount in SAFEs. This amount lowers the Series A threshold in Case 1 and raises it in Case 2. The reform cannot liquidate a firm which had a seed round, and it gives a seed round to firms which were liquidated only if the budget β was what stopped the seed investor. Figure 4 shows the two cases.

Write $\bar{l}^b$ and $\bar{l}^a$ for the largest shock with a seed round before and after the reform, $\bar{l}$ of Proposition 12 at $(\sigma_a, m) = (\sigma_a^0, 0)$ and at $(\sigma_a^0 + m, m)$. Since $\bar{l} - \sigma_a$ does not depend on σ_a and is weakly decreasing in m, $\bar{l}^a \leq \bar{l}^b + m$. A firm with $x > m$ which had a seed round keeps it, so $\bar{l}^a \geq \bar{l}^b$. The interval of shocks with a seed round moves from $(\sigma_a^0, \bar{l}^b]$ to $(\sigma_a^0 + m, \bar{l}^a]$: its lower bound rises by m and its upper bound by at most m. The probability of a seed round therefore rises if and only if $G(\bar{l}^a) - G(\bar{l}^b) > G(\sigma_a^0 + m) - G(\sigma_a^0)$, which requires more firms at the top of the interval than at its bottom.

Proposition 14. *$\bar{l}^b \leq \bar{l}^a \leq \bar{l}^b + m$, and the reform:*

(i) raises the probability that a firm needs no seed round from $G(\sigma_a^0)$ to $G(\sigma_a^0 + m)$;

(ii) does not raise the probability of a seed round if g is weakly decreasing on $[\sigma_a^0, \bar{l}^a]$;

(iii) does not raise the probability that a firm is liquidated.

→ Proof: Appendix, page 52

Integrating over shocks, the reform changes the probability of reaching Series A by $\int \left(F(t^b(l)) - F(t^a(l))\right) dG(l)$, where $t^b(l)$ and $t^a(l)$ are the thresholds before and after the reform, equal to 1 for a liquidated firm. The sign of this aggregate effect depends on G: the reform raises the threshold of the firms with $x \leq 0$ and of the firms with

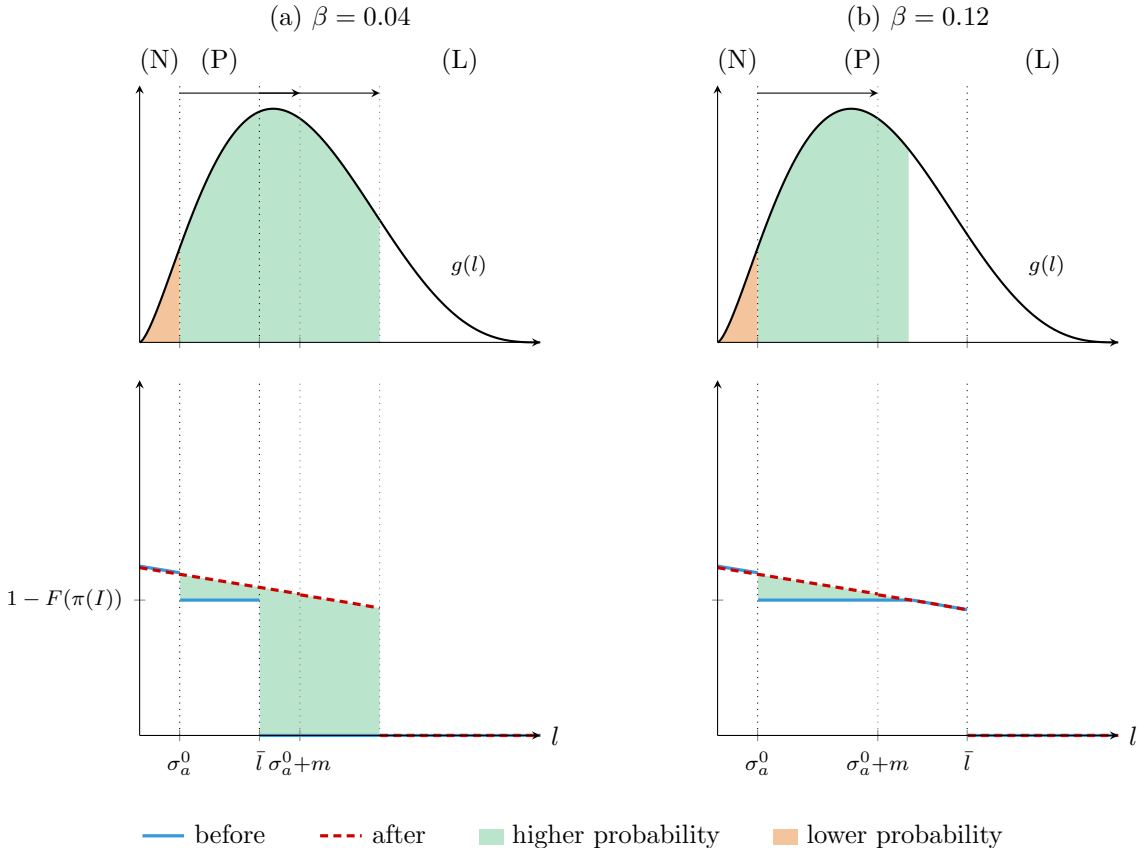


Figure 4: The probability of reaching Series A before and after the reform, by liquidity shock, when the seed investor holds a capped SAFE without discount (Proposition 13). The project's probability of success is uniform on $[0, 1]$, $I = 0.5$, $c_e = 0.005$, $\rho_a^0 = 0.07$, $\sigma_a^0 = 0.02$ and $m = 0.06$; the shock l follows a Beta$(2.5, 4)$ distribution on $[0, 0.2]$, whose density g is in the top panels.

$x > m$ under (10), it lowers the threshold of the firms in Case 1 after the reform, and it moves the threshold of the others either way.

5.6 Equity round or capped SAFE

We compare the two contracts at the same cash σ_a and the same MFN SAFE $m \geq 0$, the capped SAFE having a discount $\lambda \in (0, 1]$. At the amount σ_s, the claims of the SAFE block are worth at most $(1 - \rho_a^0)W(\sigma_A)$, their value at the best threshold when Case 2 does not constrain the cap, and the seed investor holds the fraction $\sigma_s/(\sigma_s + m)$ of them. With the equity round its claims are worth $\sigma_s/\big((1-\rho_a^0)\sigma_s + m\big)$ times $(1-\rho_a^0)W(\sigma_A)$ (Proposition 6): the equity round dilutes the accelerator's share, which replaces $\sigma_s + m$ by $(1 - \rho_a^0)\sigma_s + m$. Without the MFN SAFE, $m = 0$, the seed investor covers the need exactly under both contracts (Propositions 2 and 11), and a firm with a seed round has the threshold $\pi(I)$ with the equity round and $\max\big\{\pi(I),\, I + \big(c_e + (l - \sigma_a)/\lambda\big)/(1 - \rho_a^0)\big\}$ with the capped SAFE.

Proposition 15. *(i) At every amount σ_s, the seed investor earns at least as much with the equity round, strictly under* (5)*.*

(ii) The seed investor participates with the equity round at every shock at which it participates with the capped SAFE.

(iii) With $m = 0$, every firm's probability of reaching Series A is at least as high with the equity round.

→ Proof: Appendix, page 53

The equity round gives the seed investor more for two reasons. It dilutes the accelerator's share, which the post-money SAFE does not, so part of the seed investor's share comes from the accelerator rather than from the entrepreneur. It converts the MFN SAFE at the seed price, so the threshold of Case 2 does not constrain the seed investor. Without the MFN SAFE, the firms gain too: the seed investor covers the need exactly under both contracts, and the equity round removes the threshold of Case 2. With the MFN SAFE, the equity round exposes the firms to the loss of their seed round under the reform (Proposition 9).

5.7 Cash and claim

The reform gives the accelerator's firms m more in cash and gives the accelerator a claim of m on them. We split it in two steps. The cash step raises the accelerator's cash from σ_a^0 to $\sigma_a^0 + m$ without the MFN SAFE. The claim step then gives the accelerator the MFN SAFE m at the cash $\sigma_a^0 + m$. The seed investor raises an equity round or uses a capped SAFE with a discount $\lambda \in (0, 1]$.

Proposition 16. *Under either seed contract:*

(i) The cash step does not lower any firm's probability of reaching Series A, and it lowers the probability of liquidation from $1 - G(\bar{l})$ *to* $1 - G(\bar{l} + m)$.

(ii) The claim step raises the Series A threshold of every firm without a seed round, $l \leq \sigma_a^0 + m$*, by* $m/(1 - \rho_a^0)$*, and it does not raise the largest shock with a seed round.*

(iii) Every firm whose probability of reaching Series A falls under the reform has a lower probability after the reform than after the cash step alone.

→ Proof: Appendix, page 54

The losers of the reform come from the claim, not from the cash. The claim raises the threshold of the firms without a seed round, because the MFN SAFE converts at the Series A price and does not dilute the accelerator's share, and it narrows the set of shocks with a seed round. Writing $\mathbb{P}^b$, $\mathbb{P}^c$ and $\mathbb{P}^a$ for the probability of reaching Series A before the reform, after the cash step and after the reform,

$$\mathbb{P}^a - \mathbb{P}^b \;=\; \underbrace{\mathbb{P}^c - \mathbb{P}^b}_{\geq 0} + \big(\mathbb{P}^a - \mathbb{P}^c\big).$$

6 The design of the capped SAFE

Section 5 takes the discount λ as given. Seed investors use three designs: a cap without a discount, a discount without a cap, and a cap with a discount set by the market.

6.1 A cap without a discount

Proposition 17. *For each* σ_s*, the seed investor's profit is weakly increasing in* λ*. The seed investor proposes* $\lambda = 1$.

→ Proof: Appendix, page 55

A discount lets the marginal projects convert below the cap, where the SAFE claims are the fixed amount $(m + \sigma_s)/\lambda$. A lower λ raises the threshold $\pi_{\min}^d$ of Case 2 and narrows the set of thresholds the seed investor can reach with its cap, so the seed investor sets no discount.

6.2 A discount without a cap

A SAFE with a discount and no cap converts at $\lambda(\pi - \sigma_A)$ on every financed project, so its claims are worth $(m + \sigma_s)/\lambda$ on each of them, whatever π. An uncapped convertible note with the interest $1/\lambda - 1$ pays the same. The two differ only in seniority when the firm fails, and failure is worth zero in the model. The entrepreneur obtains $(1 - \rho_a^0)(\pi - \sigma_A) - (m + \sigma_s)/\lambda$, so the threshold is $\pi^d_{\min}$ of Section 5, and the claims are worth $(m + \sigma_s)/\lambda$ times $1 - F(\pi^d_{\min})$. A cap at $\lambda(\pi^d_{\min} - \sigma_A)$ converts the marginal project at the same price, and every project above it at a lower price than the discount.

Proposition 18. *The cap $V_s = \lambda(\pi^d_{\min} - \sigma_A)$ sets the threshold $\pi^d_{\min}$ of the SAFE with the discount λ and no cap, and every financed project converts at it. The SAFE claims are worth*

$$\frac{m + \sigma_s}{\lambda} \frac{N(\pi^d_{\min}; \sigma_A)}{\pi^d_{\min} - \sigma_A} \;\geq\; \frac{m + \sigma_s}{\lambda}\big(1 - F(\pi^d_{\min})\big)$$

with the cap and with the discount respectively, strictly if $F(\pi^d_{\min}) < 1$. The seed investor earns more with the cap.

→ Proof: Appendix, page 55

The claim at the cap is proportional to the pre-money valuation $\pi - \sigma_A$: it takes more on the good projects and less on the marginal one.

6.3 A cap with a market discount

In practice the discount is often set by the market, for instance $\lambda = 0.8$: the seed investor takes it as given and chooses its amount and its cap. Propositions 10 to 14, 15 and 16 hold for every $\lambda \in (0, 1]$, so they describe this case too. The discount enters through the slope $1/(\lambda(1-\rho_a^0)) - 1$ of the threshold of Case 2 in the amount: a lower λ raises $\pi^d_{\min}$ at every amount, lowers the bound of (10), so that more firms are in Case 2 before the reform, and makes the threshold (11) of a firm in Case 2 rise faster when the reform raises its amount (Proposition 13(iii)). In Figure 5, the discount $\lambda = 0.8$ creates losers among the firms with a seed round, which a cap without discount does not.

7 Discussion

What the Series A round stands for. The Series A round of the model is the first round at which the project's probability of success is known to the market. Investors then compete to invest in the startups that prove worth it, and none of

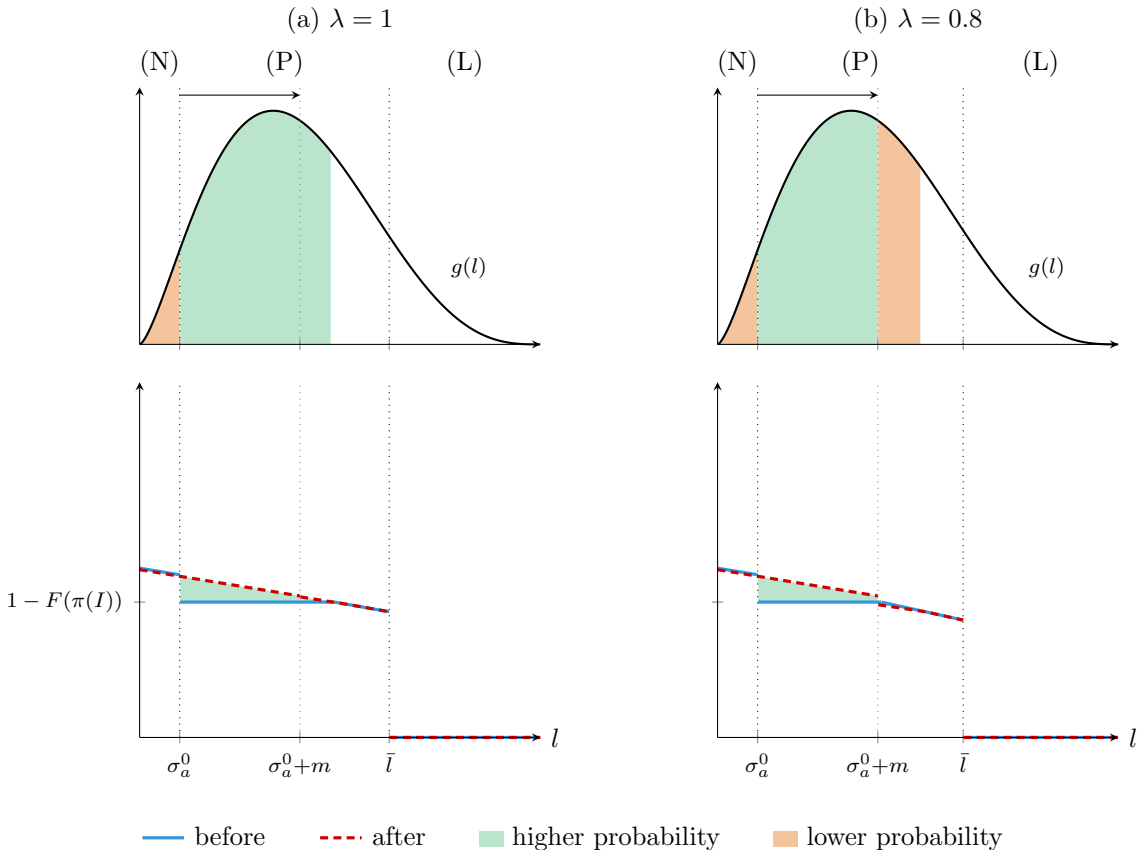


Figure 5: The probability of reaching Series A before and after the reform, by liquidity shock, when the seed investor holds a capped SAFE, without discount and with the market discount $\lambda = 0.8$; the budget is $\beta = 0.12$. The project's probability of success is uniform on $[0, 1]$, $I = 0.5$, $c_e = 0.005$, $\rho_a^0 = 0.07$, $\sigma_a^0 = 0.02$ and $m = 0.06$; the shock l follows a Beta$(2.5, 4)$ distribution on $[0, 0.2]$, whose density g is in the top panels.

them has an information advantage. In practice this round need not be called Series A: for many startups the quality of the project is only known at a Series B or a Series C. The seed investor of the model then stands for the investors who finance the firm between the accelerator and that round. More generally, the model's three investors stand for the stages of venture financing: the accelerator for the programs that select startups, the seed investor for the investors who finance the experiment under market power, and the Series A investor for the competitive rounds that follow once quality is known, up to growth rounds; the outcome stands for an acquisition or an IPO.

Failure and outside options. The outcome of a failed project and the entrepreneur's outside option in liquidation are both normalized to zero, so investors recover nothing when a project fails or is liquidated. This leaves aside the ways in which investors recover money after a failure, such as liquidation preferences, the repayment of convertible notes or the sale of the firm's assets. We do not treat these questions here.

Why seed investors use SAFEs. In the model a seed equity round gives the seed investor at least as much as a capped SAFE at every amount (Proposition 15). What makes seed investors use SAFEs is the size of an equity round. A SAFE or a convertible note converts only at an equity round above a minimum amount, a qualified financing: Techstars' former note, for instance, converted at a round of at least \$250,000.[10] A seed investor whose budget β is below this minimum cannot raise an equity round, and a SAFE is the contract left to it. In the model, the limited budget is therefore what rules out the equity round, and lower legal costs and faster closings (Coyle and Green, 2014) reinforce it.

How seed investors set the cap. The seed investor needs the distribution of outcomes, the entrepreneur's effort cost and the terms on the cap table, not the quality of a given project. In Case 1 its cap amounts to a target ownership for the SAFE block, $(m+\sigma_s)/V_s^* = 1-\rho_a^0 - c_e/\big(\pi(\sigma_A)-\sigma_A\big)$. The post-money SAFE makes this share explicit: each holder's stake is its amount over the cap, so the block's stake is the sum of the amounts over the cap. The cap rises with the block when the accelerator's MFN SAFE joins it, and the threshold of Case 1 stays at $\pi(\sigma_A)$. The cap cannot prevent the threshold of Case 2 from rising with the block, because the claims that convert at the discount do not dilute the accelerator.

The accelerator's capped SAFE. From the Winter 2019 batch, Y Combinator's 7% was a post-money SAFE with a valuation cap (YC blog, 2018). Below its cap, such a SAFE gives the accelerator the fixed claim σ_a instead of the share ρ_a^0, and the entrepreneur must be left more on the marginal project. Appendix C shows that,

[10] Techstars deal two-pager: an equity financing "resulting in aggregate sales proceeds to the Company of at least \$250,000". See also Techstars investment terms 2024.

before the reform, every firm's probability of reaching Series A is at least as high with a fixed share as with this capped SAFE (Corollary 2).

8 Conclusion

We model the financing of a startup as an experiment followed by scaling, financed by an accelerator, a seed investor and Series A investors, and we use the model to evaluate the MFN SAFE that Y Combinator added to its standard deal in 2022.

We summarize. First, with equity rounds only, the share the seed investor takes decides which projects reach Series A through the entrepreneur's effort, the seed investor covers the need exactly, and more accelerator cash lowers no firm's probability of reaching Series A. Second, Y Combinator's reform lowers the probability of reaching Series A of the firms without a seed round whatever the seed contract, and with a seed equity round it can liquidate firms that had a seed round. Third, without the clause the seed investor covers the need exactly, whether it has market power or not, and with a capped seed SAFE the reform makes seed investors invest more, and the firms whose threshold is set by the seed investor's share do not lose, but the firms whose threshold is set by the SAFE's discount lose when they receive more in SAFEs. Fourth, the seed investor's best SAFE has a cap and no discount, and a discount set by the market adds to the firms that lose. With competitive seed investors, the reform harms the firms that need no seed round and those that raise a seed equity round, and not those that use a capped SAFE.

For an accelerator, the clause has a cost for its firms whatever the seed contract: the firms that need no seed round lose, because the MFN SAFE does not dilute the accelerator's share while the Series A round it replaces does. With the accelerator's share of 7%, this loss is small, about 0.075 times the MFN SAFE on the Series A threshold. The larger effects come from the firms whose need the new cash covers, which no longer need a seed round, and, with a seed equity round, from the firms that lose their seed round.

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A Competitive seed investors

So far the seed investor has market power over the entrepreneur. We now let seed investors compete after the shock: the entrepreneur accepts the offer that gives the entrepreneur the highest expected payoff among the offers that give the seed investor a non-negative profit, and chooses the amount within the budget. Series A investors still compete. The comparison separates what the reform does through the dilution of the accelerator's share from what it does through the seed investor's market power.

A.1 Equity rounds only

Fix the amount σ_s and the Series A round σ_A. As with a monopolistic seed investor, choosing the share is choosing the threshold x, and the seed investor's claims are worth the first term of (4). The entrepreneur keeps $c_e/(x-\sigma_A)$ of the net value of each financed project, so the entrepreneur's expected payoff is

$$\frac{c_e}{x-\sigma_A}\,N(x;\sigma_A) - c_e\big(1-F(x)\big) \;=\; \frac{c_e}{x-\sigma_A}\int_x^1 \big(1-F(t)\big)\,dt,$$

which falls with the threshold. The entrepreneur therefore picks the lowest threshold at which the seed investor breaks even, the smallest x with

$$\Big(1-\frac{c_e}{(1-\rho_a)(x-\sigma_A)}\Big)N(x;\sigma_A) \;=\; \sigma_s. \tag{12}$$

The left-hand side is largest at $\pi(\sigma_A)$, where it equals $W(\sigma_A)$, so a root exists if and only if $W(\sigma_A)\geq\sigma_s$, that is if a monopolistic seed investor would take part with this amount.

Proposition 19. *With competitive seed investors and equity rounds only, the Series A investor finances the projects above the smallest root of* (12)*, which is at most the threshold $\pi(\sigma_A)$ of a monopolistic seed investor, strictly if the monopolist would earn a positive profit. The seed investor earns zero. At every need, the entrepreneur chooses the amount that covers the need exactly.*

→ Proof: Appendix, page 56

The monopolist picks the threshold at which the value of its claims is largest. Competition gives the entrepreneur the lowest threshold at which the seed investor breaks even: more projects reach Series A, and the seed investor's profit goes to the entrepreneur. Competition does not change the set of shocks with a seed round. As with a monopolistic seed investor, cash beyond the need is lost on the projects that do not reach Series A, and the entrepreneur does not ask for it.

A.2 The reform

With the MFN SAFE, the seed investor uses a capped SAFE with a discount $\lambda \in (0, 1]$, as in Section 5, or raises an equity round, as in Section 4. Read (12) with ρ_a^0 in place of ρ_a.

With the capped SAFE, a cap that converts every financed project sets a threshold $x > \pi_{\min}^d$, and the block's claims are then worth $\big(1 - \rho_a^0 - c_e/(x - \sigma_A)\big)N(x; \sigma_A)$ (Section 5). The seed investor holds the fraction $\sigma_s/(m + \sigma_s)$ of them, so it breaks even if and only if they are worth at least $m + \sigma_s$. The entrepreneur again picks the lowest threshold at which the seed investor breaks even, and no cap sets a threshold below $\pi_{\min}^d$.

Proposition 20. *With a capped SAFE and competitive seed investors, a seed round with the amount σ_s takes place if and only if a monopolistic seed investor would take part with this amount. The Series A investor then finances the projects above the larger of $\pi_{\min}^d$ and the smallest x with*

$$\Big(1 - \rho_a^0 - \frac{c_e}{x - \sigma_A}\Big)N(x; \sigma_A) \;=\; m + \sigma_s.$$

The threshold, the value of the SAFE claims and the entrepreneur's payoff depend on σ_s and m only through $m + \sigma_s$.

→ Proof: Appendix, page 57

We keep x, π_0 and the three groups of shocks of Section 4.4. The proof of Lemma 2 uses only that the seed investor's profit is non-negative, so a firm with $x \le m$ has no seed round after the reform, as with a monopolistic seed investor. For $x > m$, write S for the amount placed beyond the former cash, σ_s before the reform and $m + \sigma_s$ after it. The entrepreneur chooses S in $[x, \beta]$ before the reform and in $[x, \beta + m]$ after it. With the capped SAFE, the threshold and the entrepreneur's payoff at each S do not change (Proposition 20). With the equity round, the MFN SAFE converts at the seed price and the seed investor breaks even if and only if the left-hand side of (12) is at least $\sigma_s + m/(1 - \rho_a^0)$: at the amount S, the target is S before the reform and $S + \rho_a^0 m/(1 - \rho_a^0)$ after it.

Proposition 21. *With competitive seed investors, the reform changes the outcome of a firm as follows.*

(i) For $x \le 0$, under both contracts, the firm has no seed round before or after the reform, and its threshold rises from π_0 to $\pi_0 + \rho_a^0 m/(1 - \rho_a^0)$.

(ii) For $0 < x \le m$, under both contracts, the firm has no seed round after the reform, and its threshold is $\pi_0 + \rho_a^0 m/(1 - \rho_a^0)$.

(iii) For $x > m$, with the capped SAFE, the entrepreneur's payoff does not fall, a firm with a seed round keeps it, and its threshold does not change if the amount chosen after the reform is at most β.

(iv) For $x > m$, with the equity round, at each amount S the threshold is the smallest root of (12) *with the right-hand side S before the reform and $S + \rho_a^0 m/(1-\rho_a^0)$ after it. The threshold is strictly higher and the entrepreneur's payoff strictly lower at each amount, and the seed round takes place at fewer amounts.*

→ Proof: Appendix, page 58

The firms that need no seed round lose under both contracts: their MFN SAFE converts at the Series A price and does not dilute the accelerator's share, while the Series A round it replaces did. Among the firms that keep a seed round, the reform harms those that raise an equity round, for the same reason: the MFN SAFE converts at the seed price without diluting the accelerator, while the seed investor's cash dilutes it. It does not harm those that use a capped SAFE. The MFN SAFE then replaces seed money that did not dilute the accelerator either, and the reform only relaxes the budget. With a monopolistic seed investor, the capped SAFE gives the seed investor only part of the block, which changes the amount it places and creates the losers of Case 2 (Proposition 13).

B Proofs

The proofs use the following shorthands. $k := 1 - \rho_a$ in Section 3 and $k := 1 - \rho_a^0$ afterwards; $S := m + \sigma_s$ for the SAFE block, $S = \sigma_s$ when $m = 0$; $b := l - \sigma_a$ for the need and $\bar{b} := \bar{l} - \sigma_a$. In the proofs on the reform, S is the amount placed beyond the former cash, σ_s before the reform and $m + \sigma_s$ after it, and $\mu := \rho_a^0 m/k$.

Lemma 1. *Under each seed contract of Sections 3 to 6, a seed investor who invests less than the need, $\sigma_s \in (0,\, l - \sigma_a)$, earns $-\sigma_s < 0$, and no project reaches Series A. The seed investor therefore invests nothing or at least the need, $\sigma_s \geq l - \sigma_a$.*

← Back to the text, page 12

Proof. If $\sigma_s < l - \sigma_a$, the firm's cash $\sigma_a + \sigma_s$ does not cover the shock, and the firm is liquidated before the Series A round. No project reaches Series A and the outcome is 0 on every project, so every claim of the seed investor, a share fixed at the seed round or a SAFE, is worth 0. Its profit is $-\sigma_s < 0$, below the zero profit it earns by not investing. A seed investor who invests a positive amount therefore invests at least $l - \sigma_a$. □

Proposition 1. *Under* (5)*, equation* (6) *has a unique solution $\pi(\sigma_A)$ in $(\sigma_A, 1)$. The seed investor sets the threshold $\pi(\sigma_A)$, with the valuation*

$$V_s(\sigma_s, \sigma_A) = \frac{\sigma_s}{1 - \dfrac{c_e}{(1-\rho_a)\big(\pi(\sigma_A) - \sigma_A\big)}}.$$

← Back to the text, page 12

Proof. Write $k := 1 - \rho_a$. By the derivation before the proposition, the seed investor chooses the threshold x. A threshold $x \geq 1$ finances no project and gives $-\sigma_s$. A threshold $x \in (\sigma_A + c_e/k, 1)$, which exists under (5), gives $\varphi(x) - \sigma_s$ with

$$\varphi(x) \;:=\; \Big(1 - \frac{c_e}{k\,(x - \sigma_A)}\Big)\, N(x; \sigma_A), \tag{13}$$

which is positive since $N(x; \sigma_A) > 0$ for $x < 1$. It is enough to find the maximum of φ.

We show that φ' has the sign of $c_e - \psi$ for an increasing function ψ. Write $y := x - \sigma_A$ and $N := N(x; \sigma_A)$. Since $\partial N/\partial x = -y f(x)$,

$$\varphi'(x) \;=\; \frac{c_e\, N}{k\, y^2} - \Big(1 - \frac{c_e}{k\, y}\Big)\, y f(x) \;=\; \frac{c_e\big(N + y^2 f(x)\big) - k\, y^3 f(x)}{k\, y^2}.$$

The numerator vanishes exactly at the solutions of (6). Dividing it by $N+y^2f(x)>0$, $\varphi'(x)$ has the sign of $c_e-\psi(x)$, with

$$\psi(x) \;:=\; \frac{k\,y^3 f(x)}{y^2 f(x)+N}, \qquad x\in(\sigma_A,1). \tag{14}$$

Dividing numerator and denominator by $y^2f(x)$,

$$\psi(x) \;=\; \frac{k\,y}{1+N/\big(y^2f(x)\big)}, \tag{15}$$

so $0<\psi(x)<k\,y$.

It remains to show that ψ is strictly increasing. Write $D:=y^2f+N$. Using $\partial N/\partial x=-yf$, $\big(y^3f\big)'=3y^2f+y^3f'$ and $D'=yf+y^2f'$,

$$\frac{\psi'(x)\,D^2}{k} \;=\; \big(3y^2f+y^3f'\big)\big(y^2f+N\big)-y^3f\big(yf+y^2f'\big) \;=\; y^2\Big[\,2y^2f^2+N\big(3f+yf'\big)\Big],$$

the terms in ff' cancelling, all functions evaluated at x. Write $c:=-f'(x)/f(x)$. The bracket equals $f\big(2y^2f+N(3-cy)\big)$, so $\psi'>0$ if and only if

$$2y^2f(x) \;+\; N\big(3-c\,y\big) \;>\; 0.$$

If $c\le 3/y$, both terms are non-negative and the first is positive. If $c>3/y$, then $c>0$, and log-concavity of f in (A3) puts $\log f$ below its tangent at x: $f(t)\le f(x)e^{-c(t-x)}$ for $t\ge x$. Since $t-\sigma_A\ge 0$ on $[x,1]$,

$$N \;=\; \int_x^1 (t-\sigma_A)\,f(t)\,dt \;\le\; f(x)\int_x^\infty (t-\sigma_A)\,e^{-c(t-x)}\,dt \;=\; f(x)\,\frac{c\,y+1}{c^2},$$

and since $3-cy<0$,

$$2y^2f+N\big(3-cy\big) \;\ge\; \frac{f}{c^2}\Big(2c^2y^2+(cy+1)(3-cy)\Big) \;=\; \frac{f}{c^2}\Big((cy+1)^2+2\Big) \;>\; 0.$$

We show that ψ runs from 0 to $k(1-\sigma_A)$. As $x\to\sigma_A^+$, $0<\psi(x)<k\,y$ gives $\psi\to 0$. As $x\to 1^-$, write $r:=(1-F)/f$. By (A3), r is decreasing and non-negative, so it has a limit $L\ge 0$ at 1^-. If $L>0$, then $r\ge L$ on $(0,1)$, so $-\big(\log(1-F)\big)'=1/r\le 1/L$, and integrating from any $x_0<1$ gives $1-F(x)\ge\big(1-F(x_0)\big)e^{-(x-x_0)/L}$ for every $x<1$, which contradicts $F(1^-)=1$. Hence $L=0$. Since $t-\sigma_A\le 1-\sigma_A$ on $[x,1]$, $N\le(1-\sigma_A)\big(1-F(x)\big)$ and

$$0 \;\le\; \frac{N}{y^2f(x)} \;\le\; \frac{(1-\sigma_A)\,r(x)}{(x-\sigma_A)^2} \;\xrightarrow[x\to 1^-]{}\; 0.$$

By (15), $\psi(1^-)=k(1-\sigma_A)$.

We conclude. ψ is continuous and strictly increasing from 0 to $k(1-\sigma_A)$, and $c_e < k(1-\sigma_A)$ by (5), so $\psi(x) = c_e$, that is (6), has a unique solution $\pi(\sigma_A)$ in $(\sigma_A, 1)$. At this point $c_e = \psi < k\,y$, so $\pi(\sigma_A) > \sigma_A + c_e/k$: the threshold is admissible and the share it requires lies in $(0,1)$. φ strictly increases below $\pi(\sigma_A)$ and strictly decreases above it, so $\pi(\sigma_A)$ maximizes φ, and $\varphi\big(\pi(\sigma_A)\big) > 0$ beats every threshold $x \geq 1$. The share is $\rho_s = 1 - c_e/\big(k(\pi(\sigma_A) - \sigma_A)\big)$, and $V_s = \sigma_s/\rho_s$.

For later use, write $\rho_s^*(\sigma_A) := 1 - c_e/\big(k(\pi(\sigma_A) - \sigma_A)\big)$ for this share, so that $W(\sigma_A) = \varphi\big(\pi(\sigma_A)\big) = \rho_s^*(\sigma_A)\, N\big(\pi(\sigma_A); \sigma_A\big) > 0$ under (5). □

Corollary 1. *$\pi(\cdot)$ is strictly increasing, and $\pi(\sigma_A) > \sigma_A + c_e/(1-\rho_a)$.*

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Proof. The inequality $\pi(\sigma_A) > \sigma_A + c_e/(1-\rho_a)$ is shown in the proof of Proposition 1. Write $k := 1 - \rho_a$, and for $x > s$ write

$$\varphi_x(x; s) \;:=\; \frac{c_e\, N(x; s)}{k\,(x-s)^2} - \Big(x - s - \frac{c_e}{k}\Big) f(x)$$

for the derivative computed in that proof, with the Series A round s made explicit. By that proof, for each $s < 1 - c_e/k$ and each $x \in (s, 1)$, $\varphi_x(x; s) > 0$ if and only if $x < \pi(s)$.

Fix x and differentiate in s on $(-\infty, x)$. Since $\partial N(x; s)/\partial s = -\big(1 - F(x)\big)$,

$$\frac{\partial \varphi_x(x; s)}{\partial s} \;=\; \frac{c_e}{k}\Big(\frac{2\,N(x; s)}{(x-s)^3} - \frac{1 - F(x)}{(x-s)^2}\Big) + f(x) \;\geq\; \frac{c_e}{k}\,\frac{1 - F(x)}{(x-s)^2} + f(x) \;>\; 0,$$

where the inequality uses $N(x; s) \geq (x-s)\big(1 - F(x)\big)$, since $\pi - s \geq x - s$ for $\pi \geq x$.

Let $s < s' < 1 - c_e/k$. If $\pi(s) \leq s'$, then $\pi(s') > s' \geq \pi(s)$. Otherwise $x := \pi(s) \in (s', 1)$, and $\varphi_x(x; s') > \varphi_x(x; s) = 0$, so $x < \pi(s')$. In both cases $\pi(s') > \pi(s)$. □

Proposition 2. *A participating seed investor covers the need exactly, $\sigma_s^* = l - \sigma_a$: the Series A round is I, and the Series A investor finances the projects above $\pi(I)$.*

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Proof. Write $k := 1 - \rho_a$, $b := l - \sigma_a > 0$ for the need and $\Delta := \sigma_s - b$ for the amount invested beyond it, $\Delta \geq 0$ by Lemma 1. By Proposition 1, at the amount σ_s the Series A round is $\sigma_A = I - \Delta$, and the seed investor earns

$$W(I - \Delta) - b - \Delta \;=:\; h(\Delta) - b, \qquad \Delta \in [0, \beta - b], \tag{16}$$

with W of the proof of Proposition 1.

The marginal dollar. We show that W is continuous and decreasing, and that on $(-\infty, 1 - c_e/k)$ it is differentiable with $W' = -\Lambda$ for the function Λ defined below. Write $\bar{s} := 1 - c_e/k$. Fix $\sigma_A < \bar{s}$ and write $\Phi(x, s) := \psi(x; s) - c_e$, with ψ of (14) and the Series A round s made explicit. Φ is continuously differentiable on $\{(x, s) : s < x < 1\}$ because f is continuously differentiable by (A3), and $\partial\Phi/\partial x = \psi' > 0$ by the proof of Proposition 1. By the implicit function theorem, $\pi(\cdot)$, defined by $\Phi(\pi(s), s) = 0$, is continuously differentiable on $(-\infty, \bar{s})$.

By the proof of Proposition 1, $W(s) = \varphi\big(\pi(s); s\big)$, with φ of (13) and the Series A round s made explicit. Since $\partial\varphi/\partial x = 0$ at $x = \pi(s)$, $W'(s) = \partial\varphi/\partial s$ evaluated at $x = \pi(s)$. With $y = x - s$ and $\partial N/\partial s = -(1 - F(x))$,

$$\frac{\partial \varphi}{\partial s} \;=\; -\frac{c_e\, N}{k\, y^2} - \Big(1 - \frac{c_e}{k\, y}\Big)\big(1 - F(x)\big).$$

At $x = \pi(s)$ the first-order condition $\varphi' = 0$ of the proof of Proposition 1 reads $c_e N/(ky^2) = (1 - c_e/(ky))\, y f(x) = \rho_s^*(s)\, y f(x)$, hence

$$W'(s) \;=\; -\Lambda(s), \qquad \Lambda(s) \;:=\; \rho_s^*(s)\Big(1 - F\big(\pi(s)\big) + \big(\pi(s) - s\big) f\big(\pi(s)\big)\Big).$$

$\Lambda(s) > 0$ because $\rho_s^*(s) > 0$ (Corollary 1) and $f > 0$, so W is strictly decreasing on $(-\infty, \bar{s})$.

Continuity at $\bar{s}$. $W \geq 0$ on $(-\infty, \bar{s})$. As $s \to \bar{s}^-$, $\pi(s) \in (s + c_e/k, 1)$ and $s + c_e/k \to 1$, so $\pi(s) \to 1$ and $0 < W(s) \leq N(\pi(s); s) \leq \int_{\pi(s)}^1 f \to 0$. Hence W is continuous at $\bar{s}$, where it takes the value 0, and constant beyond.

The amount. h is continuous, so the set of its maximizers on $[0, \beta - b]$ is non-empty and closed, and its largest element Δ^* exists; $\sigma_s^* = b + \Delta^*$. Where $W(I - \Delta) > 0$, $h'(\Delta) = \Lambda(I - \Delta) - 1$; where $W(I - \Delta) = 0$, $h'(\Delta) = -1$.

Exact funding. Write $a := c_e/k$ and $R(x) := \int_x^1 \big(1 - F(t)\big)\, dt$, positive and strictly decreasing on $(0, 1)$, and recall $N(x; \sigma_A) = (x - \sigma_A)\big(1 - F(x)\big) + R(x)$. Let the seed investor place $\Delta \geq 0$ beyond the need and set the threshold $x \in (0, 1)$ through its share (proof of Proposition 1). With $u := x - I$ and $y := x - \sigma_A = u + \Delta > a$, it earns $P(x, \Delta) - b$ with

$$P(x, \Delta) \;:=\; \Big(1 - \frac{a}{y}\Big) N(x; I - \Delta) - \Delta \;=\; (u - a)\big(1 - F(x)\big) + R(x) - \Delta F(x) - \frac{a\, R(x)}{u + \Delta}.$$

A threshold $x \geq 1$ gives $-\Delta - b < 0$. Without prefunding the seed investor earns at most $W(I) - b$, with $W(I) = \max_x P(x, 0)$. We show that for every $x \in (0, 1)$ and $\Delta > 0$, either $P(x, \Delta) \leq 0$ or $P(x, \Delta) < P(x + \Delta, 0)$ with $x + \Delta < 1$. A participating seed investor earns a non-negative profit, so its gross profit is at least $b > 0$, and it then sets $\Delta = 0$.

We show that moving a dollar from the amount to the threshold raises P wherever $P > 0$. Fix $y > a$ and write $r := 1 - a/y \in (0,1)$. Along the pairs (x, Δ) with $x - I + \Delta = y$, the Series A round is $x - y$ and $P = r\,N(x; x-y) - (y - x + I)$. Since $\partial N(x; x-y)/\partial x = -y f(x) - \big(1 - F(x)\big)$,

$$\frac{dP}{dx} \;=\; 1 - r\Big(1 - F(x) + y\,f(x)\Big).$$

Since $x + R(x) = 1 - \int_x^1 F = \mathbb{E}(\pi) + \int_0^x F$,

$$P \;=\; r\,y\big(1 - F(x)\big) - y + \mathbb{E}(\pi) - I + \int_0^x F(t)\,dt - \frac{a}{y}R(x).$$

If $P > 0$, then, with $y(1-r) = a$ and $\mathbb{E}(\pi) < I + c_e \le I + a$ by (A1),

$$\int_0^x F(t)\,dt \;>\; a + r\,y\,F(x) + I - \mathbb{E}(\pi) + \frac{a}{y}R(x) \;>\; r\,y\,F(x).$$

By (A3), F is log-concave, so $F(t) \le F(x)\,e^{(t-x)f(x)/F(x)}$ for $t \le x$ and $\int_0^x F < F(x)^2/f(x)$. Hence $r\,y\,f(x) < F(x)$, and

$$\frac{dP}{dx} \;>\; 1 - r\big(1 - F(x)\big) - F(x) \;=\; (1-r)\big(1 - F(x)\big) \;\ge\; 0.$$

We conclude. Let $x \in (0,1)$ and $\Delta > 0$ with $P(x,\Delta) > 0$, and follow the pairs $(x + t, \Delta - t)$, $t \in [0, \Delta]$, which keep y. Along them P is continuous and increases wherever it is positive, so it stays above $P(x,\Delta) > 0$. At a threshold 1, $N = 0$ and $P = -(\Delta - t) \le 0$, so the path stays in $(0,1)$, and $P(x+\Delta, 0) > P(x,\Delta)$, with the positive share r. Hence every $\Delta > 0$ gives $P(x,\Delta) \le 0$ or $P(x,\Delta) < W(I)$, and a participating seed investor covers the need exactly. It earns $W(I) - b \ge 0$ with $b > 0$, so $W(I) > 0$, $I < \bar{s}$, and by Proposition 1 the Series A investor finances the projects above $\pi(I)$. □

Lemma 2. *A firm with $l \le \sigma_a$ has no seed round: the entrepreneur turns down every seed offer that gives the seed investor a non-negative profit. This holds under each seed contract of Sections 3 to 6.*

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Proof. Write k for one minus the accelerator's share ($1 - \rho_a$ in Section 3, $1 - \rho_a^0$ in the following sections), m for its MFN SAFE ($m = 0$ in Section 3), $S_0 := I + l - \sigma_a$ and $h_0(\pi) := k(\pi - S_0) - c_e$, strictly increasing.

Without a seed round. The Series A round is $\sigma_A = S_0$. The MFN SAFE converts at the Series A price, so its claims are worth m on each financed project, and the

entrepreneur obtains $h_0(\pi) - m$; the Series A investor finances the projects with $h_0(\pi) \geq m$. Let $p := \min\{\pi_0, 1\}$, where $h_0(\pi_0) = m$; $p > 0$ since $h_0(0) < 0 \leq m$. The entrepreneur's expected payoff is

$$E_N \;=\; \int_p^1 \big(h_0(\pi) - m\big) f(\pi)\, d\pi.$$

With a seed round. Let the seed investor invest $\sigma_s > 0$, so that $\sigma_A = S_0 - \sigma_s$, write $v := \pi - \sigma_A$, and let t be the Series A threshold. On a financed project $k\,v - c_e = h_0(\pi) + k\sigma_s$.

With an equity round (Sections 3 and 4), write $q = 1/V_s$. The entrepreneur holds $k - (k\sigma_s + m)q$ before the Series A round and obtains $h_0(\pi) + k\sigma_s - (k\sigma_s + m)q\,v$. The seed investor's profit is $\sigma_s\big(q\int_t^1 v f\, d\pi - 1\big)$; if it is non-negative, $(k\sigma_s + m)\, q \int_t^1 v f\, d\pi \geq k\sigma_s + m$.

With a capped SAFE (Sections 5 and 6.3), write $c(v) = \min\{V_s, \lambda v\}$. The entrepreneur obtains $h_0(\pi) + k\sigma_s - (m + \sigma_s)\, v/c(v)$. The seed investor's profit is $\sigma_s\big(\int_t^1 \frac{v}{c(v)} f\, d\pi - 1\big)$; if it is non-negative, $(m + \sigma_s)\int_t^1 \frac{v}{c(v)} f\, d\pi \geq m + \sigma_s \geq m + k\sigma_s$.

In both cases, if the seed investor's profit is non-negative, the entrepreneur's expected payoff satisfies

$$E_s \;\leq\; \int_t^1 h_0 f\, d\pi + k\sigma_s\big(1 - F(t)\big) - (m + k\sigma_s) \;=\; \int_t^1 h_0 f\, d\pi - m - k\sigma_s F(t).$$

Hence

$$E_s - E_N \;\leq\; \int_t^1 h_0 f\, d\pi - \int_p^1 h_0 f\, d\pi - mF(p) - k\sigma_s F(t).$$

Since $h_0 < m$ on $[0, p)$ and $h_0 \geq m$ on $[p, 1]$: if $t \leq p$, $\int_t^p h_0 f\, d\pi \leq m\big(F(p) - F(t)\big)$; if $t > p$, $-\int_p^t h_0 f\, d\pi \leq -m\big(F(t) - F(p)\big)$. In both cases $E_s - E_N \leq -(m + k\sigma_s)F(t)$, which is negative if $t > 0$. If $t = 0$, then $t < p$ and the first inequality is strict, since $f > 0$ on $(0, 1)$ by (A3). In both cases $E_s < E_N$. Since $l \leq \sigma_a$, the entrepreneur who turns down the offer obtains E_N, and turns it down. □

Proposition 3. *The liquidity shock sorts projects into three regimes:*

(N) No seed round, $l \leq \sigma_a$*: the Series A investor finances the projects* $\pi \geq \pi_0(l)$.

(P) Seed round, $\sigma_a < l \leq \bar{l}$*: the Series A investor finances the projects* $\pi \geq \pi(\sigma_A^*)$.

(L) Liquidation, $l > \bar{l}$*: the firm is liquidated before Series A.*

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Proof. Regime (N), $l \leq \sigma_a$. By Lemma 2 there is no seed round, so $\rho_s = 0$. The firm keeps $\sigma_a - l$, so $\sigma_A = I + l - \sigma_a$, and (3) gives the threshold $\pi_0(l)$.

Participation, $l > \sigma_a$. Let $b = l - \sigma_a > 0$. By Lemma 1, a seed round needs an amount $\sigma_s \in [b, \beta]$, hence $b \leq \beta$. The entrepreneur accepts any offer, since turning it down liquidates the firm and the offer leaves him a non-negative payoff on every financed project by (IC_e). By (16) the seed investor's largest profit at the need b is $\Pi^*(b) := \max_{[0,\beta-b]} h - b$, and it participates if and only if $\Pi^*(b) \geq 0$. h is continuous (proof of Proposition 2), so $U \mapsto \max_{[0,U]} h$ is continuous and increasing, and Π^* is continuous and strictly decreasing on $[0, \beta]$. $\Pi^*(0) \geq h(0) = W(I) \geq 0$, so with $\bar{b} := \max\{b \in [0, \beta] : \Pi^*(b) \geq 0\}$ the seed investor participates if and only if $0 < l - \sigma_a \leq \bar{b}$. Π^* does not involve σ_a, so neither does $\bar{b}$.

Regimes (P) and (L). On $(\sigma_a, \bar{l}\,]$, Proposition 1 gives the threshold $\pi(\sigma_A^*)$ at the seed investor's amount. Beyond $\bar{l}$ no seed round occurs, the shock is not covered and the firm is liquidated before π is revealed.

The largest shock. By the proof of Proposition 2, every amount beyond the need gives the seed investor a gross profit below $W(I)$ or non-positive, so $\Pi^*(b) \geq 0$ if and only if $W(I) \geq b$. Hence $\bar{l} - \sigma_a = \bar{b} = \min\{\beta, W(I)\}$.

□

Proposition 4. *The increase:*

(i) raises the probability that a firm needs no seed round from $G(\sigma_a)$ to $G(\sigma_a')$;

(ii) raises the probability of a seed round if and only if $G(\bar{l}') - G(\bar{l}) > G(\sigma_a') - G(\sigma_a)$;

(iii) lowers the probability that a firm is liquidated from $1 - G(\bar{l})$ to $1 - G(\bar{l}')$.

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Proof. By Proposition 3 the three regimes are the events $l \leq \sigma_a$, $\sigma_a < l \leq \bar{l}$ and $l > \bar{l}$, which gives the three probabilities. By the same proposition, $\bar{l} - \sigma_a = \bar{b}$ does not depend on σ_a, so $\bar{l}' = \bar{l} + \sigma_a' - \sigma_a$. The probability of no seed round goes from $G(\sigma_a)$ to $G(\sigma_a')$, which is (i); the probability of liquidation goes from $1 - G(\bar{l})$ to $1 - G(\bar{l}')$, which is (iii). The probability of a seed round goes from $G(\bar{l}) - G(\sigma_a)$ to $G(\bar{l}') - G(\sigma_a')$, and it rises if and only if $G(\bar{l}') - G(\bar{l}) > G(\sigma_a') - G(\sigma_a)$, which is (ii). □

Proposition 5. *The increase has the following effects:*

(i) no firm's probability of reaching Series A falls;

(ii) the probability of a firm with $l \le \sigma'_a$ rises if and only if $\pi'_0(l) < 1$, and the probability of a firm with $\max\{\sigma'_a, \bar{l}\} < l \le \bar{l}'$ rises;

(iii) the probability of every other firm does not change.

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Proof. Write $k := 1 - \rho_a$. F is strictly increasing on $[0,1]$ since $f > 0$ on $(0,1)$. By Proposition 2 the seed investor covers the need exactly at both amounts, so the threshold of regime (P) is $\pi(I)$, and by Proposition 3 $\bar{b}$ does not depend on the accelerator's amount: $\bar{l} = \sigma_a + \bar{b}$ and $\bar{l}' = \sigma'_a + \bar{b}$. If $\bar{b} > 0$, the seed investor's claims have a positive value, so I satisfies (5) and $\pi(I) < 1$.

Firms with $l \le \sigma_a$. They have no seed round before or after the increase, and their threshold falls from $\pi_0(l)$ to $\pi'_0(l)$. Their probability rises if and only if $\pi'_0(l) < 1$, and does not change otherwise.

Firms with $\sigma_a < l \le \sigma'_a$. After the increase they have no seed round and the threshold $\pi'_0(l) \le I + c_e/k$. Before it, they had a seed round with the threshold $\pi(I)$, if $l \le \bar{l}$, or they were liquidated. If $\pi'_0(l) < 1$, their probability rises: from zero, or from $1 - F(\pi(I))$, since $\pi(I) > I + c_e/k \ge \pi'_0(l)$ by Corollary 1. If $\pi'_0(l) \ge 1$, then $I + c_e/k \ge 1$, so $\bar{b} = 0$, the firm was liquidated, and its probability stays at zero.

Firms with $l > \sigma'_a$. If $l \le \bar{l}$, they have a seed round before and after the increase, with the threshold $\pi(I)$, and their probability does not change. If $\bar{l} < l \le \bar{l}'$, they were liquidated and have a seed round after the increase, with the threshold $\pi(I) < 1$: their probability rises. If $l > \bar{l}'$, they are liquidated before and after.

The three groups give parts (i) to (iii). □

Proposition 6. *Under* (5)*, the seed investor sets the threshold $\pi(\sigma_A)$ and proposes*

$$V_s^* \;=\; \frac{\big((1-\rho_a^0)\sigma_s + m\big)\big(\pi(\sigma_A) - \sigma_A\big)}{(1-\rho_a^0)\big(\pi(\sigma_A) - \sigma_A\big) - c_e}.$$

It earns

$$\frac{(1-\rho_a^0)\sigma_s}{(1-\rho_a^0)\sigma_s + m}\, W(\sigma_A) - \sigma_s.$$

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Proof. Write $q := 1/V_s$. The entrepreneur holds $e = k - (k\sigma_s + m)q$ before the Series A round, so choosing $q \in \big(0, k/(k\sigma_s + m)\big)$ is choosing $e \in (0,k)$, and by (IC_e) the

threshold $x = \sigma_A + c_e/e$, one to one. The seed investor holds $\sigma_s q = \frac{k\sigma_s}{k\sigma_s+m}\big(1 - \frac{e}{k}\big)$, diluted by the Series A round, so it earns

$$\frac{k\sigma_s}{k\sigma_s+m}\Big(1 - \frac{c_e}{k(x-\sigma_A)}\Big)N(x;\sigma_A) - \sigma_s \;=\; \frac{k\sigma_s}{k\sigma_s+m}\,\varphi(x) - \sigma_s,$$

with φ of (13). By Proposition 1, φ is maximal at $x = \pi(\sigma_A)$, where it is worth $W(\sigma_A)$ and $1 - e/k = \rho_s^*(\sigma_A)$, so $(k\sigma_s + m)q = k\rho_s^*(\sigma_A) = k - c_e/\big(\pi(\sigma_A) - \sigma_A\big)$, which gives V_s^*. At this valuation the seed investor earns $\frac{k\sigma_s}{k\sigma_s+m}\,W(\sigma_A) - \sigma_s$, and it earns $-\sigma_s$ if $\sigma_A \geq 1 - c_e/k$, where no project is financed. □

Proposition 7. *Under* (7)*, there is* $b_0 > 0$ *such that the seed investor invests beyond the need at every need* $l - \sigma_a \in (0, b_0)$*.*

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Proof. By the proof of Proposition 6, at the need b the seed investor earns $\frac{k\sigma_s}{k\sigma_s+m}W(I + b - \sigma_s) - \sigma_s$. By the proof of Proposition 2, W is differentiable at $I < 1 - c_e/k$ with $W' = -\Lambda$, so the right derivative of the profit at $\sigma_s = b$ is

$$\frac{km}{(kb+m)^2}\,W(I) + \frac{kb}{kb+m}\,\Lambda(I) - 1 \;\geq\; \frac{km}{(kb+m)^2}\,W(I) - 1,$$

which tends to $kW(I)/m - 1$ as $b \to 0$. By (7), the limit is positive. Hence there is $b_0 > 0$ such that, for every $b \in (0, b_0)$, the profit is strictly increasing at $\sigma_s = b$, and b is not a maximizer. □

Proposition 8. *There is* $\bar{l}$*, with* $\bar{l} - \sigma_a \in [0, \beta]$ *depending on* f*,* I*,* c_e*,* ρ_a^0*,* β *and* m *but not on* σ_a*, and weakly decreasing in* m*, such that the liquidity shock sorts projects into three regimes:*

(N) No seed round*,* $l \leq \sigma_a$*: the Series A investor finances the projects* $\pi \geq \pi_0^m(l)$*.*

(P) Seed round*,* $\sigma_a < l \leq \bar{l}$*: the Series A investor finances the projects* $\pi \geq \pi(\sigma_A^*)$*.*

(L) Liquidation*,* $l > \bar{l}$*: the firm is liquidated before Series A.*

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Proof. Regime (N), $l \leq \sigma_a$*.* There is no seed round (Lemma 2), and no equity round precedes the Series A round, so the MFN SAFE converts at the Series A price and its claims are worth m on each financed project. The firm keeps $\sigma_a - l$, so

$\sigma_A = I + l - \sigma_a$, the entrepreneur obtains $k(\pi - \sigma_A) - m$, and (IC_e) gives the threshold $\sigma_A + (c_e + m)/k = \pi_0^m(l)$.

Participation, $l > \sigma_a$. Let $b = l - \sigma_a > 0$. A seed round needs an amount $\sigma_s \in [b, \beta]$ (Lemma 1), the entrepreneur accepts any offer since turning it down liquidates the firm, and the seed investor participates if and only if $P_p^*(b) \geq 0$, where $P_p^*(b)$ is the maximum over $\sigma_s \in [b, \beta]$ of $\frac{k\sigma_s}{k\sigma_s + m} W(I + b - \sigma_s) - \sigma_s$ (proof of Proposition 6). At a given amount this profit is decreasing in b, since W is decreasing (proof of Proposition 2), and in m, since $W \geq 0$; the set of amounts shrinks when b rises; and the profit is continuous. Hence P_p^* is continuous and decreasing on $(0, \beta]$, and with $\bar{b} := \sup\{b \in (0, \beta] : P_p^*(b) \geq 0\}$, $\bar{b} = 0$ if the set is empty, the seed investor participates if and only if $0 < l - \sigma_a \leq \bar{b}$. $\bar{b}$ is decreasing in m, and since the profit depends on l and σ_a only through b, it does not depend on σ_a.

Regimes (P) and (L). On $(\sigma_a, \bar{l}\,]$, Proposition 6 gives the threshold $\pi(\sigma_A^*)$. Beyond $\bar{l}$ the shock is not covered and the firm is liquidated. □

Proposition 9. *The reform changes the probability of reaching Series A as follows.*

(i) For $x \leq 0$, it falls if $\pi_0 < 1$.

(ii) For $0 < x \leq m$, it falls if and only if the firm had a seed round and $x > \pi(I) - I - (c_e + \rho_a^0 m)/(1 - \rho_a^0)$.

(iii) For $x > m$, it falls to zero for the firms with $W(I) - \rho_a^0 m/(1 - \rho_a^0) < x \leq \min\{\beta, W(I)\}$, which lose their seed round, and it does not fall for the others. The firms with $\beta < x \leq \min\{\beta + m,\, W(I) - \rho_a^0 m/(1 - \rho_a^0)\}$ obtain a seed round.

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Proof. Write $\Psi(S) := W(I + x - S) - S$ and $\theta(S) := k(S - m)/\big(k(S - m) + m\big)$. F is strictly increasing on $[0, 1]$, and $\mu > 0$.

We show that after the reform the seed investor earns $\theta(S)\big(\Psi(S) - \mu\big)$ at the amount $S > m$. It invests $\sigma_s = S - m$, the Series A round is $I + x - S$, and by Proposition 6 it earns $\theta(S)\, W(I + x - S) - (S - m)$. Since

$$\theta(S)\big(S + \mu\big) \;=\; \frac{k(S - m)}{k(S - m) + m} \cdot \frac{kS - km + m}{k} \;=\; S - m,$$

this profit is $\theta(S)\big(\Psi(S) - \mu\big)$.

We show that, for $x > m$, the seed investor participates after the reform if and only if (8) holds. The need is $x - m > 0$, so the amounts are $S \in [x, \beta + m]$ (Lemma 1), where $\theta(S) > 0$: the seed investor participates if and only if $\Psi(S) \geq \mu$ for some such S.

With $\Delta := S - x$, $\Psi(S) + x = W(I - \Delta) - \Delta$ is the largest gross profit of a benchmark seed investor who places Δ beyond the need, and the proof of Proposition 2 shows that for $\Delta > 0$ it is below $W(I)$ or non-positive, whatever the budget. Hence every $S > x$ gives $\Psi(S) < W(I) - x = \Psi(x)$ or $\Psi(S) \leq -x < \mu$, and the seed investor participates if and only if $x \leq \beta + m$ and $W(I) - x \geq \mu$.

Part (i). There is no seed round before or after the reform (Lemma 2), and the threshold goes from π_0 to $\pi_0 + \mu$. If $\pi_0 < 1$, F rises strictly between the two and the probability falls.

Part (ii). After the reform $l \leq \sigma_a^0 + m$: there is no seed round and the threshold is $\pi_0 + \mu$. A liquidated firm had probability zero. A firm with a seed round had $0 < x \leq W(I)$, so $\pi(I) < 1$, and its probability falls if and only if $\pi(I) < \pi_0 + \mu = I + x + (c_e + \rho_a^0 m)/k$.

Part (iii). A firm with a seed round has $x \leq \min\{\beta, W(I)\}$ and the threshold $\pi(I) < 1$. If $x \leq W(I) - \mu$, (8) holds since $x \leq \beta$, and the firm keeps a seed round with $S \geq x$ and the threshold $\pi(I + x - S) \leq \pi(I)$ by Corollary 1: its probability does not fall. Otherwise the seed investor does not participate, the shock exceeds the accelerator's cash since $x > m$, and the firm is liquidated. A liquidated firm had probability zero, and it obtains a seed round if and only if (8) holds; since $x > \min\{\beta, W(I)\}$ and $x < W(I)$, this requires $x > \beta$. □

Proposition 10. *Under* (5)*, the seed investor sets the threshold* x^**, the larger of* $\pi(\sigma_A)$ *and* $\pi^d_{\min}$*, with the cap*

$$V_s^* \;=\; \frac{(m + \sigma_s)(x^* - \sigma_A)}{(1 - \rho_a^0)(x^* - \sigma_A) - c_e},$$

which equals $(\lambda c_e + m + \sigma_s)/(1 - \rho_a^0)$ *in Case 2. Every financed project converts at the cap.*

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Proof. Fix σ_s, so that S and σ_A are fixed, and write $v := \pi - \sigma_A$. When the threshold is x and every financed project converts at the cap, the block holds the share $k - c_e/(x - \sigma_A)$ of the firm before the Series A round, and its claims are worth

$$\Omega(x; \sigma_A) \;:=\; \Big(k - \frac{c_e}{x - \sigma_A}\Big)\, N(x; \sigma_A).$$

Two facts from Section 3 are used, read with ρ_a^0 in place of ρ_a. First, $\Omega(x; \sigma_A) = k\,\varphi(x)$ with φ of (13), so by the proof of Proposition 1, $\Omega(\cdot\,; \sigma_A)$ strictly increases on $(\sigma_A, \pi(\sigma_A)]$ and strictly decreases on $[\pi(\sigma_A), 1)$, with the maximum $k\,W(\sigma_A)$. Second, $\Omega(x; \sigma_A) \to 0$ as $x \to 1^-$; we set $\Omega(x; \sigma_A) := 0$ for $x \geq 1$, the value of the claims when no project is financed.

On a project with $v > 0$ the entrepreneur obtains $E(v)$, with $E(v) = (k - S/V_s)\, v$ if $v \geq V_s/\lambda$ (the project converts at the cap) and $E(v) = kv - S/\lambda$ if $v < V_s/\lambda$ (it converts at the discount); E is continuous at $v = V_s/\lambda$.

Caps $V_s \leq S/k$. Then $E(v) \leq 0$ on the cap part, and $E(v) < (kV_s - S)/\lambda \leq 0$ on the discount part. Since $c_e > 0$, no project is financed and the seed investor earns $-\sigma_s$.

Caps $V_s > S/k$. E is continuous and strictly increasing, with slopes k and $k - S/V_s$, and $E(0) = -S/\lambda < 0$, so the projects financed are those above a threshold. Write $\hat{V} := (\lambda c_e + S)/k$. Since $E(V_s/\lambda) = (kV_s - S)/\lambda$, $E(V_s/\lambda) \geq c_e$ if and only if $V_s \geq \hat{V}$.

(a) If $S/k < V_s < \hat{V}$, the threshold lies in the cap part: $x = \sigma_A + c_e/(k - S/V_s)$, and every financed project converts at the cap. The block holds $S/V_s = k - c_e/(x - \sigma_A)$ of the firm before the Series A round, so its claims are worth $\Omega(x; \sigma_A)$. The map $V_s \mapsto x$ is a strictly decreasing bijection from $(S/k, \hat{V})$ onto $(\pi^d_{\min}, \infty)$: $x \to \infty$ as $V_s \to S/k$, and at $V_s = \hat{V}$, $k - S/\hat{V} = k\lambda c_e/(\lambda c_e + S)$ and $x = \sigma_A + (c_e + S/\lambda)/k = \pi^d_{\min}$.

(b) If $V_s \geq \hat{V}$, the threshold lies in the discount part, where $E(v) = c_e$ gives $v = (c_e + S/\lambda)/k$: the threshold is $\pi^d_{\min}$ whatever V_s. On each financed project the block's claims are worth $S\, v/\min\{V_s, \lambda v\}$, which is decreasing in V_s. At $V_s = \hat{V}$, every financed project has $\lambda v \geq \lambda(\pi^d_{\min} - \sigma_A) = \hat{V}$ and converts at the cap, and the claims are worth $\frac{S}{\hat{V}} N(\pi^d_{\min}; \sigma_A) = \Omega(\pi^d_{\min}; \sigma_A)$, since $S/\hat{V} = k - c_e/(\pi^d_{\min} - \sigma_A)$. Hence the caps $V_s > \hat{V}$ give at most $\Omega(\pi^d_{\min}; \sigma_A)$.

The seed investor holds the fraction σ_s/S of the claims, so it chooses the cap that maximizes their value. By (a) and (b), the values it can reach are $\Omega(x; \sigma_A)$ for the thresholds $x \geq \pi^d_{\min}$, each reached by a single cap of (a) or, for $x = \pi^d_{\min}$, by $V_s = \hat{V}$.

Case 1, $\pi(\sigma_A) \geq \pi^d_{\min}$. $\Omega(\cdot\,; \sigma_A)$ is maximal on $[\pi^d_{\min}, \infty)$ at $x = \pi(\sigma_A)$, reached by the cap with $S/V_s = k - c_e/(\pi(\sigma_A) - \sigma_A)$.

Case 2, $\pi(\sigma_A) < \pi^d_{\min}$. $\Omega(\cdot\,; \sigma_A)$ is decreasing on $[\pi^d_{\min}, \infty)$, so its maximum is at $x = \pi^d_{\min}$, reached by $V_s = \hat{V}$.

In both cases every financed project converts at the cap, and the claims are worth $\Omega\big(\max\{\pi(\sigma_A), \pi^d_{\min}\}; \sigma_A\big)$. At the amount σ_s the seed investor therefore earns

$$P(\sigma_s) \;:=\; \frac{\sigma_s}{S}\, \Omega\big(\max\{\pi(\sigma_A), \pi^d_{\min}\}; \sigma_A\big) - \sigma_s,$$

with $\Omega := 0$ when no project is financed. □

Proposition 11. *(i) If $m = 0$, the seed investor covers the need exactly: $\sigma^*_s = l - \sigma_a$.*

(ii) Under (7) and (9), there is $b_0 > 0$ such that the seed investor invests beyond the need, $\sigma^*_s > l - \sigma_a$, at every need $l - \sigma_a \in (0, b_0)$.

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Proof. At the need b, $\sigma_A = I + b - \sigma_s$ and $\pi^d_{\min} = I + b - \sigma_s + \big(c_e + (m+\sigma_s)/\lambda\big)/k = I + b + (c_e + m/\lambda)/k + \kappa\sigma_s$, with $\kappa := 1/(\lambda k) - 1 > 0$. The seed investor earns $P(\sigma_s)$ of the proof of Proposition 10.

(i). Let $m = 0$, so that $S = \sigma_s$, and write $\Delta := \sigma_s - b \geq 0$ and $\pi^d_{\min}(\Delta) := I - \Delta + \big(c_e + (b+\Delta)/\lambda\big)/k$, which is increasing in Δ since $\lambda k \leq 1$. By the proof of Proposition 10, the seed investor earns $\max_{x \geq \pi^d_{\min}(\Delta)} \Omega(x; I-\Delta) - b - \Delta$. With P of the proof of Proposition 2, read with ρ^0_a in place of ρ_a, $\Omega(x; I-\Delta) - \Delta = k\,P(x,\Delta) - \rho^0_a \Delta \leq k\,P(x,\Delta)$ and $\Omega(x; I) = k\,P(x,0)$. Let $\Delta > 0$ and $\pi^d_{\min}(\Delta) \leq x < 1$. By that proof, either $P(x,\Delta) \leq 0$, or $P(x,\Delta) < P(x',0)$ with $x' := x + \Delta < 1$ and $x' - I > a$. Then $x' \geq \pi^d_{\min}(\Delta) \geq \pi^d_{\min}(0)$ is a threshold the seed investor can set without investing beyond the need, and it earns more there. A participating seed investor earns at least $b > 0$ gross, so it covers the need exactly.

(ii). Let $m > 0$. At $\sigma_s = b$, $\sigma_A = I$ and $\pi^d_{\min} = I + \big(c_e + (m+b)/\lambda\big)/k$, which tends to $I + (c_e + m/\lambda)/k < \pi(I)$ as $b \to 0$ by (9). By continuity there are $b_1 > 0$ and, for each $b \in (0, b_1)$, an $\varepsilon > 0$ such that Case 1 holds on $[b, b+\varepsilon)$, where $P(\sigma_s) = \frac{\sigma_s}{m+\sigma_s} kW(I + b - \sigma_s) - \sigma_s$. Its right derivative at $\sigma_s = b$ is

$$\frac{m}{(m+b)^2}\, kW(I) + \frac{b}{m+b}\, k\Lambda(I) - 1 \;\geq\; \frac{m}{(m+b)^2}\, kW(I) - 1,$$

which tends to $kW(I)/m - 1$ as $b \to 0$, positive by (7). Hence there is $b_0 \in (0, b_1]$ such that, for every $b \in (0, b_0)$, P is strictly increasing at b, and b is not a maximizer: $\sigma^*_s > b$. □

Proposition 12. *There is $\bar{l}$, with $\bar{l} - \sigma_a \in [0, \beta]$ depending on f, I, c_e, ρ^0_a, β, m and λ but not on σ_a, and weakly decreasing in m, such that the liquidity shock sorts projects into three regimes:*

- *(N)* No seed round, *$l \leq \sigma_a$: the Series A investor finances the projects $\pi \geq \pi^m_0(l)$.*
- *(P)* Seed round, *$\sigma_a < l \leq \bar{l}$: the Series A investor finances the projects $\pi \geq \max\big\{\pi(\sigma^*_A),\, \pi^d_{\min}\big\}$, evaluated at the seed investor's amount.*
- *(L)* Liquidation, *$l > \bar{l}$: the firm is liquidated before Series A.*

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Proof. Regime (N), $l \leq \sigma_a$. By Lemma 2 there is no seed round, so no SAFE is issued after the MFN SAFE, and as in the proof of Proposition 8 it converts at the Series A price and the threshold is $\pi^m_0(l)$.

Participation, $l > \sigma_a$. Let $b = l - \sigma_a > 0$. A seed round needs an amount $\sigma_s \in [b, \beta]$ (Lemma 1), the entrepreneur accepts any offer since turning it down liquidates the firm, and the seed investor participates if and only if $P^*(b) := \max_{\sigma_s \in [b,\beta]} P(\sigma_s) \geq 0$, with P of the proof of Proposition 10.

For an amount σ_s, a need b and an MFN SAFE m, write

$$\mathcal{V}(\sigma_s, b, m) \;:=\; \max_{x \geq \pi^d_{\min}} \Omega(x; \sigma_A) \;\geq\; 0,$$

with $\sigma_A = I + b - \sigma_s$ and $\pi^d_{\min} = \sigma_A + \big(c_e + (m + \sigma_s)/\lambda\big)/k$, so that $P(\sigma_s) = \frac{\sigma_s}{m+\sigma_s}\mathcal{V} - \sigma_s$ by the proof of Proposition 10.

$\mathcal{V}$ is decreasing in b and in m. $\pi^d_{\min}$ is increasing in b and in m, so the set of thresholds shrinks. σ_A rises with b. Let $s < s'$ and let x be such that $\Omega(x; s') \geq 0$, that is $k - c_e/(x - s') \geq 0$. For every $u \in [s, s']$, $k - c_e/(x - u) \geq 0$, and

$$\frac{\partial \Omega(x; u)}{\partial u} \;=\; -\frac{c_e\, N(x; u)}{(x-u)^2} - \Big(k - \frac{c_e}{x-u}\Big)\big(1 - F(x)\big) \;\leq\; 0,$$

so $\Omega(x; s') \leq \Omega(x; s)$. Since $\mathcal{V} \geq 0$, only the thresholds with $\Omega \geq 0$ matter, and $\mathcal{V}$ is decreasing in b.

The interval. At a given amount, P is decreasing in b, and the set $[b, \beta]$ of amounts shrinks when b rises, so P^* is decreasing on $(0, \beta]$. Ω is continuous and the set of thresholds moves continuously with (σ_s, b), so $\mathcal{V}$, P and P^* are continuous by the maximum theorem. Hence, with $\bar{b} := \sup\{b \in (0, \beta] : P^*(b) \geq 0\}$ and $\bar{b} = 0$ if the set is empty, $\{b \in (0, \beta] : P^*(b) \geq 0\} = (0, \bar{b}]$ when it is not empty, and the seed investor participates if and only if $0 < l - \sigma_a \leq \bar{b}$.

σ_a and m. P depends on l and σ_a only through b, so $\bar{b}$ does not depend on σ_a. $\frac{\sigma_s}{m+\sigma_s}$ is decreasing in m and $\mathcal{V} \geq 0$ is decreasing in m, so P^* and $\bar{b}$ are decreasing in m.

Regimes (P) and (L). On $(\sigma_a, \bar{l}\,]$, Proposition 10 gives the threshold $\max\{\pi(\sigma^*_A), \pi^d_{\min}\}$. Beyond $\bar{l}$ the shock is not covered and the firm is liquidated. □

Proposition 13. *The reform changes the probability of reaching Series A as follows.*

(i) For $x \leq 0$, it falls if $\pi_0 < 1$.

(ii) For $0 < x \leq m$, it falls if and only if the firm had a seed round and $\pi^b(x) < \pi_0 + \rho^0_a m/(1 - \rho^0_a)$.

(iii) For $x > m$, a firm which had a seed round keeps it. Its probability does not fall if Case 1 holds at the amount it receives after the reform. Under (10), its probability does not rise, and it falls if the seed investor invests beyond the need after the reform. The probability of a firm which was liquidated does not fall.

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Proof. F is strictly increasing on $[0,1]$ since $f>0$ on $(0,1)$, and $\mu>0$. A firm whose seed investor participates has a threshold below 1, since the seed investor's claims are worth nothing otherwise and the need is positive.

Thresholds. Before the reform $m=0$ and $\sigma_a=\sigma_a^0$; after it $\sigma_a=\sigma_a^0+m$. In both cases the block is S and $\sigma_A=I+x-S$, so

$$\pi^d_{\min} \;=\; I+x-S+\frac{c_e+S/\lambda}{k} \;=\; \pi_0+\Big(\frac{1}{\lambda k}-1\Big)S,$$

and by Proposition 10 the threshold of a firm with a seed round is $T(S)$ of (11), Case 2 holding when its second term is the larger. The first term is strictly decreasing in S by Corollary 1, and the second is strictly increasing since $\lambda k<1$. Before the reform $S=x$ by Proposition 11(i), so the threshold is $T(x)=\pi^b(x)$. A firm without a seed round has the threshold $\pi_0^m(l)$, which is π_0 before the reform and $I+x-m+(c_e+m)/k=\pi_0+\mu$ after it.

We show that a firm with $x>m$ which had a seed round keeps it, with an amount $S\geq x$. The Series A round $I+x-S$ and the threshold $\pi_0+\big(1/(\lambda k)-1\big)S$ of Case 2 depend on S only, so the value of the SAFE claims at the best cap, $\mathcal{V}(S):=\max_{z\geq\pi_0+(1/(\lambda k)-1)S}\Omega(z;I+x-S)$, depends on S only (proof of Proposition 10). Write $\Psi(S):=\mathcal{V}(S)-S$. Before the reform the seed investor holds all the claims, earns $\Psi(S)$, and participates: $\Psi(x)\geq 0$. After the reform the need is $x-m$, so it places $\sigma_s=S-m$ with $S\in[x,\beta+m]$ (Lemma 1), holds the fraction $(S-m)/S$ of the claims and earns $\frac{S-m}{S}\mathcal{V}(S)-(S-m)=\frac{S-m}{S}\Psi(S)$. At $S=x$ this profit is non-negative since $x>m$, so the seed investor participates, with $S\geq x$.

Part (i). $l\leq\sigma_a^0\leq\sigma_a^0+m$, so by Lemma 2 there is no seed round before or after the reform, and the threshold goes from π_0 to $\pi_0+\mu$. If $\pi_0<1$, F rises strictly between the two and the probability falls.

Part (ii). After the reform $l\leq\sigma_a^0+m$: there is no seed round and the threshold is $\pi_0+\mu$. A liquidated firm had probability zero. A firm with a seed round had the threshold $\pi^b(x)<1$, and its probability falls if and only if $\pi^b(x)<\pi_0+\mu$.

Part (iii). By the previous step the firm keeps its seed round, and its threshold goes from $T(x)$ to $T(S)$ with $S\geq x$. Case 1 holds exactly on the amounts below the one at which the two terms of T are equal. If Case 1 holds at S, it holds on $[x,S]$, where T is decreasing, so $T(S)\leq T(x)$. Under (10), Case 2 holds at x, hence on $[x,S]$, where T is strictly increasing, so $T(S)\geq T(x)$, strictly if $S>x$, and the probability then falls since $T(x)<1$. A liquidated firm had probability zero. □

Proposition 14. $\bar{l}^b\leq\bar{l}^a\leq\bar{l}^b+m$, *and the reform:*

(i) raises the probability that a firm needs no seed round from $G(\sigma_a^0)$ to $G(\sigma_a^0 + m)$;

(ii) does not raise the probability of a seed round if g is weakly decreasing on $[\sigma_a^0, \bar{l}^a]$;

(iii) does not raise the probability that a firm is liquidated.

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Proof. By Proposition 12, $\bar{b} := \bar{l} - \sigma_a$ does not depend on σ_a and is weakly decreasing in m, and a firm needs no seed round if and only if $l \leq \sigma_a$, has a seed round if and only if $\sigma_a < l \leq \bar{l}$, and is liquidated otherwise.

We show that $\bar{l}^b \leq \bar{l}^a \leq \bar{l}^b + m$. $\bar{l}^a - \sigma_a^0 - m = \bar{b}(m) \leq \bar{b}(0) = \bar{l}^b - \sigma_a^0$. If $\bar{l}^b \leq \sigma_a^0 + m$, then $\bar{l}^b \leq \bar{l}^a$ since $\bar{l}^a \geq \sigma_a^0 + m$. Otherwise the set of needs with a seed round before the reform is $(0, \bar{b}(0)]$ (proof of Proposition 12), so the firm with the shock $\bar{l}^b$ has a seed round before the reform and $x = \bar{b}(0) > m$; it keeps its seed round (proof of Proposition 13), and $\bar{l}^b \leq \bar{l}^a$.

Parts (i) and (iii). The probability of no seed round goes from $G(\sigma_a^0)$ to $G(\sigma_a^0 + m)$, and the probability of liquidation from $1 - G(\bar{l}^b)$ to $1 - G(\bar{l}^a) \leq 1 - G(\bar{l}^b)$.

Part (ii). The probability of a seed round goes from $G(\bar{l}^b) - G(\sigma_a^0)$ to $G(\bar{l}^a) - G(\sigma_a^0 + m)$. Write $L := \bar{l}^b - \sigma_a^0$ and $L' := \bar{l}^a - \sigma_a^0 - m$, so that $0 \leq L' \leq L$. If g is weakly decreasing on $[\sigma_a^0, \bar{l}^a]$,

$$G(\bar{l}^a) - G(\sigma_a^0 + m) \;=\; \int_{\sigma_a^0}^{\sigma_a^0 + L'} g(t + m)\, dt \;\leq\; \int_{\sigma_a^0}^{\sigma_a^0 + L'} g(t)\, dt \;\leq\; G(\bar{l}^b) - G(\sigma_a^0).$$

□

Proposition 15. *(i) At every amount σ_s, the seed investor earns at least as much with the equity round, strictly under* (5)*.*

(ii) The seed investor participates with the equity round at every shock at which it participates with the capped SAFE.

(iii) With $m = 0$, every firm's probability of reaching Series A is at least as high with the equity round.

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Proof. Write $k = 1 - \rho_a^0$, and fix the need $b = l - \sigma_a$ and the amount σ_s, so that $\sigma_A = I + b - \sigma_s$.

Part (i). If $\sigma_A \geq 1 - c_e/k$, no project is financed under either contract and both profits are $-\sigma_s$. Otherwise, with the capped SAFE the seed investor earns

$$\frac{\sigma_s}{m + \sigma_s} \max_{x \geq \pi^d_{\min}} \Omega(x; \sigma_A) - \sigma_s$$

(proof of Proposition 10), and $\max_{x \geq \pi^d_{\min}} \Omega \leq \max_x \Omega = kW(\sigma_A)$. With the equity round it earns $\frac{k\sigma_s}{k\sigma_s + m} W(\sigma_A) - \sigma_s = \frac{\sigma_s}{k\sigma_s + m} kW(\sigma_A) - \sigma_s$ (proof of Proposition 6). Since $k\sigma_s + m < \sigma_s + m$ when $\rho^0_a > 0$, the equity round gives at least as much, and strictly more since $W(\sigma_A) > 0$ for $\sigma_A < 1 - c_e/k$ (proof of Proposition 1).

Part (ii). Both contracts allow the same amounts $[b, \beta]$, so by part (i) the seed investor's largest profit at the need b is at least as high with the equity round. If it is non-negative with the capped SAFE, it is non-negative with the equity round.

Part (iii). Let $m = 0$. A firm with $l \leq \sigma_a$ has no seed round under either contract (Lemma 2) and the same threshold $I + l - \sigma_a + c_e/k$ (Propositions 12 and 8). A firm with a seed round under the capped SAFE has one under the equity round by part (ii). Under both contracts the seed investor covers the need exactly (Propositions 2 and 11(i), the equity round with $m = 0$ being the benchmark with ρ^0_a in place of ρ_a), so $\sigma_A = I$, and the threshold is $\max\{\pi(I),\, I + (c_e + b/\lambda)/k\}$ with the capped SAFE (Proposition 10) and $\pi(I)$ with the equity round (Proposition 6). A firm liquidated under the capped SAFE has probability zero. □

Proposition 16. *Under either seed contract:*

(i) The cash step does not lower any firm's probability of reaching Series A, and it lowers the probability of liquidation from $1 - G(\bar{l})$ *to* $1 - G(\bar{l} + m)$.

(ii) The claim step raises the Series A threshold of every firm without a seed round, $l \leq \sigma^0_a + m$, *by* $m/(1 - \rho^0_a)$, *and it does not raise the largest shock with a seed round.*

(iii) Every firm whose probability of reaching Series A falls under the reform has a lower probability after the reform than after the cash step alone.

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Proof. Write $k = 1 - \rho^0_a$. Before the reform the accelerator has the cash σ^0_a and no MFN SAFE; after the cash step, the cash $\sigma^0_a + m$ and no MFN SAFE; after the reform, the cash $\sigma^0_a + m$ and the MFN SAFE m.

Without the MFN SAFE. With $m = 0$, the seed investor covers the need exactly under both contracts (Propositions 2 and 11(i)). At the cash σ_a and the need $b = l - \sigma_a \in (0, \bar{b}]$, the Series A round is therefore I, and the threshold is $\tau(b) := \pi(I)$

with an equity round (Proposition 6) and $\tau(b) := \max\{\pi(I),\, I + (c_e + b/\lambda)/k\}$ with a capped SAFE (Proposition 10 with $\sigma_A = I$ and $S = b$). In both cases τ is increasing in b, and $\tau(b) \geq \pi(I) > I + c_e/k$ by Corollary 1. A firm without a seed round has the threshold $I + l - \sigma_a + c_e/k$ (Propositions 12 and 8 with $m = 0$), and $\bar{b}$ does not depend on σ_a (Propositions 12 and 8).

Part (i). Fix l and write $b = l - \sigma_a^0$. If $b \leq 0$, the firm has no seed round before or after the cash step, and its threshold falls by m. If $0 < b \leq m$, after the cash step it has no seed round and the threshold $I + b - m + c_e/k \leq I + c_e/k < \tau(b)$; before, it had a seed round with the threshold $\tau(b)$ or it was liquidated. If $b > m$ and $b \leq \bar{b}$, the firm has a seed round before and after, with the needs b and $b - m \in (0, \bar{b}]$, and $\tau(b - m) \leq \tau(b)$. If $b > \bar{b}$, the firm was liquidated. In every case the probability of reaching Series A does not fall. The largest shock with a seed round is $\sigma_a + \bar{b}$, which rises from $\bar{l}$ to $\bar{l} + m$.

Part (ii). At the cash $\sigma_a = \sigma_a^0 + m$, a firm with $l \leq \sigma_a$ has no seed round with or without the MFN SAFE (Lemma 2). Its threshold is $\pi_0^m(l) = I + l - \sigma_a + (c_e + m)/k$ with the MFN SAFE and $I + l - \sigma_a + c_e/k$ without it (Propositions 12 and 8): it rises by m/k. By Propositions 12 and 8, $\bar{b}$ is decreasing in m, so the largest shock with a seed round, $\sigma_a + \bar{b}$, does not rise.

Part (iii). If a firm's probability is lower after the reform than before it, it is lower after the reform than after the cash step, since by part (i) the cash step does not lower it. □

Proposition 17. *For each σ_s, the seed investor's profit is weakly increasing in λ. The seed investor proposes $\lambda = 1$.*

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Proof. By the proof of Proposition 10, at a given σ_s the seed investor earns

$$\frac{\sigma_s}{S} \max_{x \geq \pi^d_{\min}} \Omega(x; \sigma_A) - \sigma_s.$$

Ω and σ_A do not depend on λ, and $\pi^d_{\min} = \sigma_A + (c_e + S/\lambda)/k$ is decreasing in λ: a larger λ enlarges the set of thresholds over which the same function is maximized. Hence the profit is increasing in λ at every σ_s, and $\lambda = 1$ is optimal on $(0, 1]$. □

Proposition 18. *The cap $V_s = \lambda(\pi^d_{\min} - \sigma_A)$ sets the threshold $\pi^d_{\min}$ of the SAFE with the discount λ and no cap, and every financed project converts at it. The SAFE claims are worth*

$$\frac{m + \sigma_s}{\lambda} \frac{N(\pi^d_{\min}; \sigma_A)}{\pi^d_{\min} - \sigma_A} \;\geq\; \frac{m + \sigma_s}{\lambda}\big(1 - F(\pi^d_{\min})\big)$$

with the cap and with the discount respectively, strictly if $F(\pi^d_{\min}) < 1$. *The seed investor earns more with the cap.*

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Proof. Write $y := \pi^d_{\min} - \sigma_A$, so that $ky - c_e = S/\lambda > 0$.

The discount. Every financed project converts at $\lambda(\pi - \sigma_A)$, so the SAFE claims are worth S/λ on it, and the entrepreneur's constraint reads $k(\pi - \sigma_A) - S/\lambda \geq c_e$, that is $\pi \geq \pi^d_{\min}$. The claims are worth $\frac{S}{\lambda}\big(1 - F(\pi^d_{\min})\big)$.

The cap. With $V_s = \lambda y$, every project with $\pi \geq \pi^d_{\min}$ has $\lambda(\pi - \sigma_A) \geq V_s$ and converts at the cap. Then $k - S/V_s = (ky - S/\lambda)/y = c_e/y$, and the constraint reads $(c_e/y)(\pi - \sigma_A) \geq c_e$, that is $\pi \geq \pi^d_{\min}$: the threshold is $\pi^d_{\min}$. The claims are worth $\frac{S}{V_s} N(\pi^d_{\min}; \sigma_A) = \frac{S}{\lambda} \frac{N(\pi^d_{\min}; \sigma_A)}{y}$.

The comparison. $N(\pi^d_{\min}; \sigma_A) - y\big(1 - F(\pi^d_{\min})\big) = \int_{\pi^d_{\min}}^1 (\pi - \pi^d_{\min}) f(\pi)\, d\pi \geq 0$, strictly if $F(\pi^d_{\min}) < 1$. The seed investor holds the fraction σ_s/S of the claims in both cases and pays σ_s, so it earns more with the cap. □

Proposition 19. *With competitive seed investors and equity rounds only, the Series A investor finances the projects above the smallest root of* (12)*, which is at most the threshold* $\pi(\sigma_A)$ *of a monopolistic seed investor, strictly if the monopolist would earn a positive profit. The seed investor earns zero. At every need, the entrepreneur chooses the amount that covers the need exactly.*

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Proof. Write $k := 1 - \rho_a$, $y := x - \sigma_A$, $R(x) := \int_x^1 \big(1 - F(t)\big)\, dt$, and φ for the function of (13), so that the left-hand side of (12) is $\varphi(x)$. Integration by parts gives $N(x; \sigma_A) = y\big(1 - F(x)\big) + R(x)$.

The entrepreneur's payoff. If the Series A investor finances the projects above x and the entrepreneur holds the share c_e/y of the net value of each financed project, the entrepreneur obtains $\frac{c_e}{y}(\pi - \sigma_A) - c_e$ on each project $\pi \geq x$, and in expectation

$$U(x) \;=\; \frac{c_e}{y} N(x; \sigma_A) - c_e\big(1 - F(x)\big) \;=\; \frac{c_e\, R(x)}{x - \sigma_A},$$

strictly decreasing in x on $(\sigma_A, 1)$ since R is strictly decreasing there.

The offer. With the share ρ_s, (IC_e) gives the threshold $x = \sigma_A + c_e/\big(k(1 - \rho_s)\big)$, one to one with $\rho_s \in (0, 1)$, and the entrepreneur holds $k(1 - \rho_s) = c_e/y$. The seed investor earns $\rho_s N(x; \sigma_A) - \sigma_s = \varphi(x) - \sigma_s$; a threshold $x \geq 1$ gives it $-\sigma_s < 0$. The

entrepreneur maximizes $U(x)$ subject to $\varphi(x) \geq \sigma_s$. By the proof of Proposition 1, φ is zero at $\sigma_A + c_e/k$, strictly increasing up to $\pi(\sigma_A)$, where it equals $W(\sigma_A)$, and strictly decreasing beyond. The constraint set is a closed interval when $\sigma_s \leq W(\sigma_A)$ and empty otherwise. The entrepreneur picks its smallest element, the smallest root of $\varphi(x) = \sigma_s$, at which the seed investor earns zero. This root is at most $\pi(\sigma_A)$, with equality only if $\sigma_s = W(\sigma_A)$.

Participation. A monopolistic seed investor takes part with the amount σ_s if and only if $W(\sigma_A) - \sigma_s \geq 0$ (proof of Proposition 2), the same condition, and earns $W(\sigma_A) - \sigma_s > 0$ exactly when the root lies strictly below $\pi(\sigma_A)$.

Exact funding. Write $b := l - \sigma_a$, $a := c_e/k$, $u := x - I$, and $P(x, \Delta)$ of the proof of Proposition 2. With the amount $b + \Delta$ and the threshold x, the Series A round is $I - \Delta$, the seed investor earns $P(x, \Delta) - b$, and the entrepreneur earns $U(x) = k\,aR(x)/(u + \Delta)$. Without investing beyond the need, the entrepreneur obtains the smallest root T_0 of $\varphi(x; I) = b$ when $b \leq W(I)$.

Suppose that some $\Delta_0 > 0$ and x_0 give the seed investor a non-negative profit, $P(x_0, \Delta_0) \geq b$, and the entrepreneur at least what exact funding gives, if any. Write $c := R(x_0)/(x_0 - I + \Delta_0)$, and for $x \geq x_0$ let $y(x) := R(x)/c$ and

$$\Phi(x) \;:=\; P\big(x,\, y(x) - x + I\big) \;=\; x - I - a + aF(x) + R(x) - F(x)\, y(x) - a\, c,$$

so that $\Phi(x_0) \geq b > 0$ and the entrepreneur's payoff along the path is constant. $y(x) - (x - I)$ is strictly decreasing, positive at x_0 and negative near 1, so it vanishes at a unique $x_c \in (x_0, 1)$. Differentiating, $\Phi'(x) = F\big(1 + (1-F)/c\big) - f\,(y - a)$. If $\Phi(x) > 0$, then since $x + R(x) = \mathbb{E}(\pi) + \int_0^x F$ and $\mathbb{E}(\pi) < I + a$ by (A1), $\int_0^x F > F(y - a)$; the log-concavity of F gives $\int_0^x F < F^2/f$ (proof of Proposition 2), so $f\,(y - a) < F$ and $\Phi'(x) > F(1 - F)/c > 0$. Hence Φ is positive and strictly increasing on $[x_0, x_c]$. At a point where $y = a > x - I$, $\Phi = x - I - a < 0$, so $y > a$ along the path and $x_c - I = y(x_c) > a$. Then $\varphi(x_c; I) = P(x_c, 0) = \Phi(x_c) > b$, so $b < W(I)$ and T_0 exists. Since $R(x)/(x - I)$ is strictly decreasing and $R(x_c)/(x_c - I) = c \geq R(T_0)/(T_0 - I)$, $x_c \leq T_0$, and since $\varphi(\cdot\,; I)$ is increasing below $T_0 \leq \pi(I)$, $\varphi(x_c; I) \leq b$, a contradiction. Every amount beyond the need therefore gives the entrepreneur strictly less than exact funding. □

Proposition 20. *With a capped SAFE and competitive seed investors, a seed round with the amount σ_s takes place if and only if a monopolistic seed investor would take part with this amount. The Series A investor then finances the projects above the larger of $\pi^d_{\min}$ and the smallest x with*

$$\Big(1 - \rho^0_a - \frac{c_e}{x - \sigma_A}\Big) N(x; \sigma_A) \;=\; m + \sigma_s.$$

The threshold, the value of the SAFE claims and the entrepreneur's payoff depend on σ_s and m only through $m + \sigma_s$.

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Proof. Write $k = 1-\rho_a^0$ and $S = m+\sigma_s$, and use Ω and $\hat{V} := (\lambda c_e + S)/k$ of the proof of Proposition 10; the left-hand side of the equation is $\Omega(x;\sigma_A)$. By that proof, a cap $V_s < \hat{V}$ sets a threshold $x > \pi^d_{\min}$, one to one, at which every financed project converts at the cap; the block's claims are worth $\Omega(x;\sigma_A)$, the entrepreneur holds $c_e/(x-\sigma_A)$ before the Series A round and obtains $U(x)$ of the proof of Proposition 19, with ρ_a^0 in place of ρ_a. A cap $V_s \geq \hat{V}$ sets the threshold $\pi^d_{\min}$; the claims are then worth $C(V_s)$, continuous and decreasing from $\Omega(\pi^d_{\min};\sigma_A)$ at $\hat{V}$, and the entrepreneur obtains $\int_{\pi^d_{\min}}^1 \big(k(\pi-\sigma_A)-c_e\big) f\, d\pi - C(V_s)$, equal to $U(\pi^d_{\min})$ at $\hat{V}$ and increasing in V_s.

The seed investor holds the fraction σ_s/S of the claims, so it breaks even if and only if they are worth at least S. If $\Omega(\pi^d_{\min};\sigma_A) \geq S$, the threshold $\pi^d_{\min}$ is feasible and the entrepreneur prefers it to every larger threshold, since U is decreasing and the caps $V_s \geq \hat{V}$ give at least $U(\pi^d_{\min})$. Otherwise the feasible thresholds are the $x > \pi^d_{\min}$ with $\Omega(x;\sigma_A) \geq S$, and the entrepreneur picks the smallest. Since $\Omega = k\varphi$ has the shape of φ, the smallest root x_S of $\Omega = S$ lies at or below $\pi^d_{\min}$ in the first case and above it in the second, so the threshold is $\max\{\pi^d_{\min}, x_S\}$. A feasible threshold exists if and only if $\max_{x\geq\pi^d_{\min}} \Omega(x;\sigma_A) \geq S$, which is the condition under which a monopolistic seed investor's profit $\frac{\sigma_s}{S}\max_{x\geq\pi^d_{\min}}\Omega - \sigma_s$ is non-negative (proof of Proposition 10).

$\pi^d_{\min}$, Ω, C, the break-even condition and the entrepreneur's payoff involve S, σ_A, λ and k, and not σ_s or m separately. □

Proposition 21. *With competitive seed investors, the reform changes the outcome of a firm as follows.*

(i) For $x \leq 0$, under both contracts, the firm has no seed round before or after the reform, and its threshold rises from π_0 to $\pi_0 + \rho_a^0 m/(1-\rho_a^0)$.

(ii) For $0 < x \leq m$, under both contracts, the firm has no seed round after the reform, and its threshold is $\pi_0 + \rho_a^0 m/(1-\rho_a^0)$.

(iii) For $x > m$, with the capped SAFE, the entrepreneur's payoff does not fall, a firm with a seed round keeps it, and its threshold does not change if the amount chosen after the reform is at most β.

(iv) For $x > m$, with the equity round, at each amount S the threshold is the smallest root of (12) with the right-hand side S before the reform and $S + \rho_a^0 m/(1-\rho_a^0)$ after it. The threshold is strictly higher and the entrepreneur's payoff strictly lower at each amount, and the seed round takes place at fewer amounts.

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Proof. Write $k = 1 - \rho_a^0$ and S for the amount placed beyond the former cash, σ_s before the reform and $m + \sigma_s$ after it, so that $\sigma_A = I + x - S$ in both cases.

Parts (i) and (ii). The proof of Lemma 2 uses only that the seed investor's profit is non-negative, so the lemma holds with competitive seed investors, and a firm with $l \leq \sigma_a$ has no seed round. Its MFN SAFE converts at the Series A price, and its threshold is $\pi_0^m(l)$ (proof of Proposition 8), which is π_0 before the reform and $\pi_0 + \rho_a^0 m/k$ after it.

Part (iii). Before the reform the amount $\sigma_s = S$ ranges over $[x, \beta]$ by Lemma 1. After it the need is $x - m > 0$ and $\sigma_s = S - m$ ranges over $[x - m, \beta]$, that is $S \in [x, \beta + m]$. By Proposition 20, the threshold, the break-even condition and the entrepreneur's payoff at each S are the same before and after the reform. The entrepreneur maximizes the same payoff over a larger set, so the payoff does not fall and a firm with a seed round keeps it; if the largest best amount after the reform lies in $[x, \beta]$, it is the largest best amount before the reform, and the threshold is the same.

Part (iv). With the seed price V_s, the entrepreneur holds $e = k(1 - \sigma_s/V_s) - m/V_s$ before the Series A round, (IC_e) gives the threshold $x = \sigma_A + c_e/e$, and the entrepreneur obtains $U(x)$ of the proof of Proposition 19. From $e = c_e/(x - \sigma_A)$, $V_s = (k\sigma_s + m)/\big(k - c_e/(x - \sigma_A)\big)$, and the seed investor breaks even if and only if $\frac{\sigma_s}{V_s} N(x; \sigma_A) \geq \sigma_s$, that is $\varphi(x) \geq \sigma_s + m/k$, with φ of (13). As in the proof of Proposition 19, the entrepreneur picks the smallest such threshold. Before the reform $m = 0$ and the target is S; after it $\sigma_s = S - m$ and the target is $S - m + m/k = S + \rho_a^0 m/k$. At a given S the Series A round is the same, and a higher target has a strictly higher smallest root on the increasing part of φ; by the monotonicity of U the entrepreneur's payoff is strictly lower, and the amounts with $W(I + x - S)$ at least the target are fewer. □

C The accelerator's capped SAFE

In the body of the paper the accelerator's share ρ_a^0 is fixed. From the Winter 2019 batch, Y Combinator's 7% was a post-money SAFE with a valuation cap (YC blog, 2018), and a capped SAFE gives at least the share it promises, more if the equity round is priced below the cap. We study this contract before the reform, $m = 0$, with a seed investor who holds a capped SAFE with a discount $\lambda \in (0, 1]$.

The accelerator invests σ_a on a post-money SAFE with the cap $V_a := \sigma_a/\rho_a^0$. At the Series A round, with the pre-money valuation $v = \pi - \sigma_A$, it converts at $\min\{V_a, v\}$, so it holds the share $\max\{\rho_a^0, \sigma_a/v\}$ of the firm before the Series A round, and its claim is worth

$$A(v) \;:=\; \max\{\rho_a^0\, v,\ \sigma_a\}.$$

Above the cap, $v \geq V_a$, the accelerator holds ρ_a^0, as with a fixed share. Below it, the accelerator's claim is the fixed amount σ_a. The SAFEs do not dilute one another, so the entrepreneur obtains $v - A(v) - S\,v/\min\{V_s, \lambda v\}$ on the project, with $S = \sigma_s$. A cap $V_s \leq S/k$ would leave the entrepreneur no share of the firm at the caps, and we consider $V_s > S/k$.

Write Ω for the value of the SAFE claims of the proof of Proposition 10. When the threshold is x and every financed project converts at the seed investor's cap, the SAFE block holds $1 - A(x - \sigma_A)/(x - \sigma_A) - c_e/(x - \sigma_A)$ of the firm before the Series A round, and its claims are worth

$$\begin{aligned}\Omega_a(x; \sigma_A) \;&:=\; \Big(1 - \frac{A(x - \sigma_A) + c_e}{x - \sigma_A}\Big) N(x; \sigma_A) \\ &=\; \min\Big\{\Omega(x; \sigma_A),\ \Big(1 - \frac{\sigma_a + c_e}{x - \sigma_A}\Big) N(x; \sigma_A)\Big\}.\end{aligned}$$

The second function is the seed investor's gross profit of the benchmark with no accelerator share and the effort cost $c_e + \sigma_a$: below the cap, the accelerator's fixed claim acts on the marginal project as an additional effort cost. Write $\tilde{\pi}(\sigma_A)$ for its maximizer, given by Proposition 1 with $\rho_a = 0$ and $c_e + \sigma_a$ in place of c_e, and

$$x_a^*(\sigma_A) \;:=\; \begin{cases} \pi(\sigma_A) & \text{if } \pi(\sigma_A) - \sigma_A \geq V_a, \\ \tilde{\pi}(\sigma_A) & \text{if } \tilde{\pi}(\sigma_A) - \sigma_A \leq V_a, \\ \sigma_A + V_a & \text{otherwise.} \end{cases}$$

Write π_a^d for the threshold at which the marginal project converts at the seed investor's discount: $\pi_a^d - \sigma_A$ solves $v - A(v) - S/\lambda = c_e$. Proposition 1 applies to the two functions in the minimum, read with $(1 - \rho_a, c_e)$ equal to $(1 - \rho_a^0, c_e)$ and to $(1, c_e + \sigma_a)$, when

$$\sigma_A \;<\; 1 - \max\Big\{\frac{c_e}{1 - \rho_a^0},\ c_e + \sigma_a\Big\}. \tag{17}$$

With a fixed share, the threshold is $T := \max\{\pi(\sigma_A), \pi_{\min}^d\}$ (Proposition 10). The cap changes the accelerator's claim only on the projects whose pre-money valuation is below V_a, so it can move the threshold only when $T - \sigma_A < V_a$.

Without a seed round, the entrepreneur obtains $v - A(v)$, which is $(1-\rho_a^0)v$ above the cap and $v - \sigma_a$ below it. The threshold is $\sigma_A + c_e/(1-\rho_a^0)$ if $c_e/(1-\rho_a^0) \geq V_a$. Otherwise the marginal project lies below the cap, and the threshold is $\sigma_A + c_e + \sigma_a$, above $\sigma_A + c_e/(1-\rho_a^0)$ since $\sigma_a = \rho_a^0 V_a > \rho_a^0 c_e/(1-\rho_a^0)$.

Proposition 22. *Under* (17)*, at each amount* σ_s*:*

(i) the seed investor proposes the cap that sets the Series A threshold at $T_a := \max\{x_a^*(\sigma_A),\, \pi_a^d\}$*, and every financed project converts at its cap;*

(ii) $T_a = T$ *if* $T - \sigma_A \geq V_a$*, and* $T_a > T$ *otherwise;*

(iii) the seed investor earns at most what it earns when the accelerator's share is fixed, strictly less if $T - \sigma_A < V_a$*.*

Proof. Write $\rho = \rho_a^0$, $k = 1 - \rho$, $v = x - \sigma_A$ for a threshold x, and, for $j \in \{k, 1\}$,

$$\Omega_k(x) \;:=\; \Big(k - \frac{c_e}{v}\Big)N(x;\sigma_A) = \Omega(x;\sigma_A), \qquad \Omega_1(x) \;:=\; \Big(1 - \frac{c_e + \sigma_a}{v}\Big)N(x;\sigma_A).$$

By the proof of Proposition 1, read with $(1-\rho_a, c_e) = (k, c_e)$ and $(1, c_e + \sigma_a)$, which applies under (17), Ω_k strictly increases below $\pi(\sigma_A)$ and strictly decreases above it, and Ω_1 does the same around $\tilde{\pi}(\sigma_A)$. Since $1 - A(v)/v = \min\{1-\rho,\, 1 - \sigma_a/v\}$ and $N > 0$, $\Omega_a = \min\{\Omega_k, \Omega_1\}$, and $\Omega_k - \Omega_1 = (\sigma_a/v - \rho)N$ has the sign of $V_a - v$: $\Omega_a = \Omega_1$ for $v \leq V_a$ and $\Omega_a = \Omega_k$ for $v \geq V_a$.

The maximizer of Ω_a*.* If $\pi(\sigma_A) - \sigma_A \geq V_a$, then $\Omega_a(\pi(\sigma_A)) = \Omega_k(\pi(\sigma_A)) = \max \Omega_k \geq \Omega_a$ everywhere. If $\tilde{\pi}(\sigma_A) - \sigma_A \leq V_a$, then likewise $\Omega_a(\tilde{\pi}(\sigma_A)) = \max \Omega_1 \geq \Omega_a$. Otherwise $\pi(\sigma_A) - \sigma_A < V_a < \tilde{\pi}(\sigma_A) - \sigma_A$: for $v < V_a$, $\Omega_a = \Omega_1$ is strictly increasing, and for $v > V_a$, $\Omega_a = \Omega_k$ is strictly decreasing. In every case Ω_a is maximal at $x_a^*(\sigma_A)$, strictly increasing below it and decreasing above it. When both of the first two conditions hold, $\Omega_k(\pi(\sigma_A)) = \Omega_a(\pi(\sigma_A)) \leq \Omega_a(\tilde{\pi}(\sigma_A)) = \Omega_1(\tilde{\pi}(\sigma_A))$ and symmetrically, so the two values coincide and the two definitions agree on the value of Ω_a at its maximum.

Part (i). The entrepreneur obtains $E(v) = v - A(v) - S\,v/\min\{V_s, \lambda v\}$. E is continuous and piecewise linear, with slopes among 1, k, $1 - S/V_s$ and $k - S/V_s$, all positive since $V_s > S/k$, so E is strictly increasing and the financed projects are those above a threshold. As in the proof of Proposition 10, a cap at which the marginal project converts at the cap sets the threshold x with $E(v) = c_e$, every financed project converts at the cap, and the block holds $S/V_s = 1 - (A(v) + c_e)/v$, so its claims are worth $\Omega_a(x)$; these thresholds are those above π_a^d, one to one. A cap at which the marginal project converts at the discount sets the threshold π_a^d, and the block's claims are worth at most $\Omega_a(\pi_a^d)$, which the cap at which the marginal project converts at both reaches. The seed investor holds the whole block, $m = 0$, so it maximizes Ω_a over $x \geq \pi_a^d$, which by the shape of Ω_a gives the threshold $\max\{x_a^*, \pi_a^d\} = T_a$, reached by a cap at which every financed project converts at the cap.

The two thresholds of Case 2. Write $v^d = \pi^d_{\min} - \sigma_A = (c_e + S/\lambda)/k$ and $v^d_a = \pi^d_a - \sigma_A$. $E_a(v) := v - A(v) - S/\lambda$ is continuous and strictly increasing, equal to $kv - S/\lambda$ for $v \geq V_a$ and to $v - \sigma_a - S/\lambda$ for $v < V_a$. If $v^d \geq V_a$, $v^d_a = v^d$. If $v^d < V_a$, then $E_a(v^d) = v^d - \sigma_a - S/\lambda < kv^d - S/\lambda = c_e$ because $\sigma_a = \rho V_a > \rho v^d$, so $v^d_a > v^d$.

Part (ii). Write $y := \pi(\sigma_A) - \sigma_A$. If $y < V_a$, then $x^*_a > \pi(\sigma_A)$: in the third case $x^*_a - \sigma_A = V_a > y$; in the second case, write $\psi_1(v)$ for the function ψ of (14) with $1 - \rho_a = 1$, strictly increasing, so that $\psi_1(\tilde{\pi}(\sigma_A) - \sigma_A) = c_e + \sigma_a$ and $\psi_1(y) = c_e/k$; since $y > c_e/k$ by Corollary 1, $\sigma_a = \rho V_a > \rho y > \rho c_e/k$, so $c_e + \sigma_a > c_e/k$ and $\tilde{\pi}(\sigma_A) > \pi(\sigma_A)$.

Let $T - \sigma_A \geq V_a$. In Case 1, $T = \pi(\sigma_A)$, $y \geq V_a$ and $x^*_a = \pi(\sigma_A)$; moreover $\pi(\sigma_A) \geq \pi^d_{\min}$ reads $ky - S/\lambda \geq c_e$, that is $E_a(y) \geq c_e$, so $\pi^d_a \leq \pi(\sigma_A)$ and $T_a = T$. In Case 2, $T = \pi^d_{\min}$ with $v^d \geq V_a$, so $\pi^d_a = \pi^d_{\min}$; and $x^*_a \leq \pi^d_{\min}$, since $x^*_a = \pi(\sigma_A) < \pi^d_{\min}$ if $y \geq V_a$ and $x^*_a - \sigma_A \leq V_a \leq v^d$ otherwise. Hence $T_a = T$.

Let $T - \sigma_A < V_a$. In Case 1, $y < V_a$ and $T_a \geq x^*_a > \pi(\sigma_A) = T$. In Case 2, $v^d < V_a$ and $T_a \geq \pi^d_a > \pi^d_{\min} = T$.

Part (iii). $\Omega_a \leq \Omega$ and $\pi^d_a \geq \pi^d_{\min}$, so $\max_{x \geq \pi^d_a} \Omega_a \leq \max_{x \geq \pi^d_{\min}} \Omega$. If $T - \sigma_A < V_a$, then $T_a \neq T$, $T_a \geq \pi^d_{\min}$, and T is the unique maximizer of Ω on $[\pi^d_{\min}, 1)$, so $\Omega_a(T_a) \leq \Omega(T_a) < \Omega(T)$. □

The cap changes nothing when the marginal project's pre-money valuation at the Series A round is above it. When it is below, the accelerator's claim on the marginal project is σ_a instead of $\rho^0_a v$, the entrepreneur must be left more, and the threshold rises. In dollars, the cap implied by YC's terms is about $2 million, so the floor binds only on projects whose pre-money valuation at the Series A round is below this amount.

With a fixed share the seed investor covers the need exactly (Proposition 11(i)). With the capped SAFE, the accelerator's claim below the cap acts on the marginal project as an additional effort cost, and the argument of Proposition 2 carries over.

Corollary 2. *Under* (17) *at $\sigma_A = I$, under both forms of the accelerator's share, the seed investor covers the need exactly. At every shock, the probability of reaching Series A is at least as high with a fixed share as with the accelerator's capped SAFE, and the largest shock with a seed round is at least as large.*

Proof. Exact funding. With a fixed share this is Proposition 11(i). With the capped SAFE, let the seed investor place $\Delta \geq 0$ beyond the need b and set the threshold x. The Series A round is $I - \Delta$, write $y := x - I + \Delta$, and the seed investor earns $H(x, \Delta) - b$ with

$$H(x, \Delta) \;:=\; \Omega_a(x; I - \Delta) - \Delta \;=\; \Big(1 - \frac{c(y)}{y}\Big) N(x; I - \Delta) - \Delta,$$

where $c(y) := A(y) + c_e \geq c_e$. At fixed y, $c(y)$ is fixed. The step "moving a dollar from the amount to the threshold" of the proof of Proposition 2, with $c(y)$ in place of a and $r := 1 - c(y)/y$, uses only $c(y) \geq c_e$, (A1) and (A3), so $dH/dx > 0$ along the pairs that keep y wherever $H > 0$. As in that proof, every $x \in (0, 1)$ and $\Delta > 0$ give $H(x, \Delta) \leq 0$ or $H(x, \Delta) < H(x + \Delta, 0)$, with $x + \Delta < 1$ and a positive share at $\Delta = 0$.

The threshold π_a^d of Case 2 is weakly increasing in Δ: v_a^d rises with slope $1/(\lambda k)$ or $1/\lambda$, at least 1, while σ_A falls with slope 1. A threshold $x \geq \pi_a^d$ set at Δ therefore gives $x + \Delta \geq \pi_a^d$ at $\Delta = 0$, a threshold the seed investor can set without investing beyond the need, with the claims $\Omega_a(x+\Delta; I)$ (proof of Proposition 22). A participating seed investor earns a gross profit of at least $b > 0$, so it covers the need exactly: $\sigma_s^* = b$.

The comparison. Under both forms the Series A round of a firm with a seed round is therefore I. By Proposition 22(iii), at every amount the seed investor earns at most what it earns with a fixed share, so its largest profit at each need is at most as high, and the largest shock with a seed round is at least as large with a fixed share. A firm with a seed round under the capped SAFE has one with a fixed share, with the threshold $T \leq T_a$ by Proposition 22(ii) at $\sigma_A = I$. A firm without a seed round has the threshold $\sigma_A + c_e/k$ with a fixed share and at least as much with the capped SAFE. A firm liquidated under the capped SAFE has probability zero. □

C.1 The capped SAFE and the MFN SAFE

The accelerator's capped SAFE and its MFN SAFE both hold the larger of a share and a fixed amount: the capped SAFE is worth $\max\{\rho_a^0 v, \sigma_a\}$, and the MFN SAFE, when the seed investor holds a capped SAFE without discount, $\max\{m\, v/V_s,\, m\}$. Both are post-money SAFEs, so neither dilutes the other SAFEs, and the Series A round dilutes both. They differ in who sets the share and where the floor binds.

	Accelerator's capped SAFE	Accelerator's MFN SAFE
Share	ρ_a^0, set by the accelerator before the shock	m/V_s, set by the seed investor's cap after the shock
Floor	σ_a	m
The floor binds when	$v < V_a$, a cap fixed in advance	$v < V_s/\lambda$, the seed investor's cap: Case 2
Without a seed round	$\max\{\rho_a^0 v, \sigma_a\}$	m: no share of the outcome
At the marginal project	raises the threshold if $T - \sigma_A < V_a$ (Proposition 22)	leaves the threshold of Case 1 at $\pi(\sigma_A)$, raises the threshold (11) of Case 2

Table 2: The accelerator's capped SAFE and its MFN SAFE, when the seed investor holds a capped SAFE.

With the MFN SAFE, the share is chosen by the seed investor, who holds part of

the block and sets the cap so that the threshold of Case 1 stays at $\pi(\sigma_A)$. With the capped SAFE, the share is chosen by the accelerator before the shock, and the seed investor takes it as given: when the floor binds at the marginal project, the seed investor cannot undo it, and the threshold rises. Without a seed round, the MFN SAFE is a fixed claim m on every financed project, while the capped SAFE keeps the share ρ_a^0 on the projects above its cap.